



OPEN ACCESS

EDITED BY
Serena Meraviglia,
University of Palermo, Italy

REVIEWED BY
Xiuli Wu,
Jinan University, China
Jingjing Pu,
Shanghai Jiao Tong University, China

*CORRESPONDENCE
Zhe Zhang
✉ z Zhang@cmu.edu.cn
Guojun Zhang
✉ gj Zhang77@cmu.edu.cn

RECEIVED 26 December 2025
REVISED 29 March 2026
ACCEPTED 08 April 2026
PUBLISHED 24 April 2026

CITATION
Zhou X, Zhai M, Zhang Z
and Zhang G (2026) Translational
potential of $\gamma\delta$ T cells in hematologic
diseases: from immunobiology to
therapeutic innovation.
Front. Immunol. 17:1775981.
doi: 10.3389/fimmu.2026.1775981

COPYRIGHT
© 2026 Zhou, Zhai, Zhang and Zhang.
This is an open-access article distributed
under the terms of the [Creative
Commons Attribution License \(CC BY\)](#).
The use, distribution or reproduction in
other forums is permitted, provided the
original author(s) and the copyright
owner(s) are credited and that the
original publication in this journal is
cited, in accordance with accepted
academic practice. No use, distribution
or reproduction is permitted which does
not comply with these terms.

Translational potential of $\gamma\delta$ T cells in hematologic diseases: from immunobiology to therapeutic innovation

Xiaokuan Zhou¹, Mengqi Zhai¹, Zhe Zhang^{2*}
and Guojun Zhang^{1*}

¹Department of Hematology, The Fourth Affiliated Hospital of China Medical University, Shenyang, Liaoning, China, ²Department of Urology, The First Affiliated Hospital of China Medical University, Shenyang, Liaoning, China

Hematologic disorders, including malignant and autoimmune conditions, present persistent clinical challenges characterized by relapse, treatment resistance, and profound immune dysregulation. While conventional immunotherapies have advanced, their efficacy is frequently limited by HLA downregulation and effector T cell exhaustion. In this context, $\gamma\delta$ T cells offer a promising therapeutic alternative. Recognizing antigens independently of MHC restriction, $\gamma\delta$ T cells possess intrinsic tissue-homing capabilities and exhibit dual cytotoxic and immunoregulatory functions. These properties make them highly suitable candidates for allogeneic, “off-the-shelf” cellular therapies where $\alpha\beta$ T cells face alloreactive limitations. This review systematically synthesizes the immunobiology of $\gamma\delta$ T cells, exploring the functional heterogeneity of specific subsets and their regulation within the tumor microenvironment (TME). We critically evaluate recent preclinical and clinical evidence supporting adoptive transfer, CAR- $\gamma\delta$ T strategies, and combination regimens across acute leukemias, lymphomas, multiple myeloma, and immune cytopenias. Furthermore, we address critical translational barriers—including *in vivo* persistence, subset exhaustion, and manufacturing variability—and discuss rational engineering strategies, metabolic preconditioning, and epigenetic modulation as solutions. Ultimately, advancing $\gamma\delta$ T cell therapies requires overcoming these hurdles to transition them effectively from the bench to mainstream clinical practice.

KEYWORDS

CAR- $\gamma\delta$ T cell therapy, cytokine plasticity, hematologic malignancies, translational barriers, $\gamma\delta$ T cells

Introduction

Hematologic disorders, both benign and malignant, represent a heavy and expanding burden on global health. Non-malignant conditions such as autoimmune hemolytic anemia (AIHA), severe aplastic anemia (SAA), and immune thrombocytopenia (ITP) are far from harmless. One-year mortality of severe aplastic anemia reaches 30–40%, while autoimmune

hemolytic anemia triples the risk of thromboembolism, with death in nearly one quarter of cases (1–3). Malignant blood system diseases are in the same situation. In elderly patients with acute myeloid leukemia (AML), the five-year survival rate falls below 20%, and relapse brings a median overall survival (OS) of only six to eight months (4, 5). Relapse itself is the familiar outcome across acute lymphoblastic leukemia (ALL), chronic lymphocytic leukemia (CLL), and multiple myeloma (MM), most severe in those with adverse genetic profiles (6–8). Even as targeted drugs and immune checkpoint inhibitors emerge, access remains profoundly unequal: in low- and middle-income countries, fewer than five percent of eligible patients undergo allogeneic stem cell transplantation (9–11). Resistance to therapy, immune evasion through HLA loss, relentless relapse, and toxic side effects all make clear the need for new immune-based treatments—therapies beyond the confines of MHC, designed for universal use.

$\gamma\delta$ T cells represent a distinct group of T lymphocytes, positioned between innate and adaptive immunity (12, 13). Conventional $\alpha\beta$ T cells, in contrast, depend on MHC-restricted recognition and fall neatly into CD4+ helper and CD8+ cytotoxic categories (12, 14). $\gamma\delta$ T cells do not follow this rule. They detect a wide variety of antigens—stress ligands, phosphoantigens, and lipids—without the need for MHC. This feature provides a potential therapeutic advantage in hematologic malignancies with HLA downregulation: it gives them an advantage against hematologic malignancies that escape immune attack by reducing HLA expression (15–17). Their behavior is fast and decisive. They kill directly, release pro-inflammatory cytokines, and move with accuracy to tissues such as the bone marrow, the site where many blood disorders begin (18, 19). Their biology looks almost paradoxical. They are innate-like, yet they adapt. They can be modified with tools like CAR constructs (20–22). Crucially, their MHC-independent recognition intrinsically bypasses alloreactivity and the risk of graft-versus-host disease (GVHD). This supports the potential of $\gamma\delta$ T cells as a promising platform for developing universal, allogeneic ‘off-the-shelf’ immunotherapies, eliminating the need for patient-specific HLA matching. Just as important, they play both sides of the immune system—driving activation when needed and imposing regulation when required. That dual role allows them to address malignancy and autoimmunity alike, making them unusually versatile across the spectrum of disease.

This review offers a translational perspective on $\gamma\delta$ T cell biology and their therapeutic promise in hematologic disease, with attention to unmet needs, subset-specific mechanisms, and evidence from early clinical trials. We bring together current findings on their cytotoxic and regulatory roles, approaches to engineering, and applications across leukemia, lymphoma, multiple myeloma, and autoimmune cytopenias. Particular focus is given to obstacles that stand in the way—functional exhaustion, inconsistency of expansion, and donor diversity—and to strategies that may overcome them, such as cytokine conditioning, bispecific targeting, and GMP-compliant manufacturing. The purpose is simple but far-reaching: to connect immunobiology with clinical necessity, and to suggest how $\gamma\delta$ T cells may be developed into a new platform of immunotherapy for patients with refractory or high-risk blood disorders.

Overview and translational implications of $\gamma\delta$ T cells

Overview of the T cell system: a prelude to $\gamma\delta$ T cell identity

T cells illustrate how the immune system achieves a balance between precise recognition and rapid response. They originate in the bone marrow and mature in the thymus, where they are shaped into distinct categories (Figure 1). The majority carry $\alpha\beta$ T cell receptors (TCRs), fitting neatly into the CD4+ helper or CD8+ cytotoxic roles, both tied to MHC restriction (23–25). Their dominance in lymphoid organs reinforces a predictable logic: recognition depends on antigen display through MHC. Yet a smaller group— $\gamma\delta$ T cells—defies this structure. They act more like intuitive responders, less dependent on complex signaling. With rapid, MHC-independent responses and strong tissue localization, they occupy a hybrid space between innate and adaptive immunity (26–28). Their very unpredictability makes them attractive candidates for immunotherapy.

Structural and functional distinctions between $\gamma\delta$ and $\alpha\beta$ T cells

The contrast between $\gamma\delta$ and $\alpha\beta$ T cells reflects two different modes of immune cognition. $\gamma\delta$ TCRs, built from gamma and delta chains, recognize stress ligands, phosphoantigens, and non-peptide targets without requiring MHC mediation (29–31). $\alpha\beta$ T cells, in contrast, rely on a slower, two-step verification: antigen presentation plus co-stimulatory signals (12, 32, 33). This design ensures accuracy but costs time. $\gamma\delta$ T cells, needing only a single trigger such as phosphoantigen binding, respond with speed. Their roles—cytotoxic attack, cytokine release, tissue repair—emerge most vividly in epithelial and mucosal sites (34–36).

Clinically, they can attack tumors or pathogens that escape $\alpha\beta$ surveillance through HLA downregulation (37–39). Their fast responses carve out a unique niche—promising in cancers where conventional $\alpha\beta$ T cells fail because the tumor has learned to hide.

Functional subsets of $\gamma\delta$ T cells and disease correlation

The map of $\gamma\delta$ T cell subsets reveals more than biological taxonomy (Figure 2); it marks possible roads for precision therapy. V γ 9V δ 2 cells, long adapted to recognize metabolic irregularities, align with the needs of anti-leukemic strategies, while V δ 1 cells, guardians of epithelial barriers, hold promise against solid tumors (Table 1).

Translational implications: $\gamma\delta$ T cells as universal therapeutic candidates

$\gamma\delta$ T cells have several biological features that make them attractive candidates for cellular immunotherapy, including MHC-independent recognition, low alloreactivity, tissue-homing capacity, and rapid effector function. Their instinctive tissue-

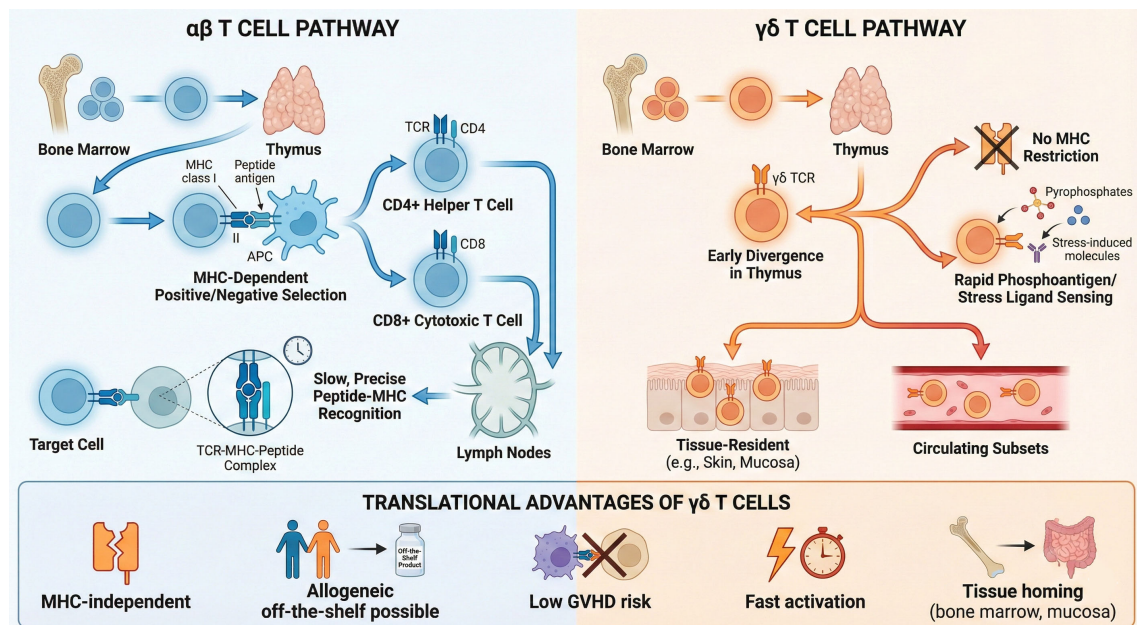


FIGURE 1

Comparative positioning of $\alpha\beta$ and $\gamma\delta$ T cells within the immune system and their translational implications. Conventional $\alpha\beta$ T cells require classical antigen-presenting cells (APCs) to process and present specific peptide antigens via Major Histocompatibility Complex (MHC) molecules. This recognition by the $\alpha\beta$ T-cell receptor (TCR) is heavily restricted by human leukocyte antigen (HLA) matching, which drives a slower, adaptive immune response and carries a high risk of alloreactivity and graft-versus-host disease (GVHD) in allogeneic transplant settings. In stark contrast, $\gamma\delta$ T cells operate independently of MHC restriction. The $\gamma\delta$ TCR, alongside innate co-receptors such as NKG2D, directly recognizes non-peptidic antigens (e.g., phosphoantigens) and stress-induced ligands (e.g., MICA/B) expressed on the surface of transformed or infected cells. This innate-like, rapid recognition mechanism intrinsically bypasses the need for patient-specific HLA matching and minimizes alloreactive GVHD risk, firmly establishing $\gamma\delta$ T cells as ideal candidates for the development of universal, “off-the-shelf” allogeneic cellular immunotherapies. APC, antigen-presenting cell; MHC, major histocompatibility complex; TCR, T-cell receptor; HLA, human leukocyte antigen; GVHD, graft-versus-host disease; NKG2D, natural killer group 2 member D; MICA/B, MHC class I chain-related protein A and B.

homing and rapid recognition capabilities make them natural candidates for therapies (40–42). Unlike $\alpha\beta$ T cells, they can operate effectively within hypoxic and immune-suppressive tumor micro-environments effectively (12, 42, 43). Even more striking is their adaptability in the laboratory: they can be multiplied, shaped, and armed with engineered receptors while retaining cytotoxic activity (44–46).

While $\gamma\delta$ T cells offer unique biological advantages that have the potential to complement existing therapies, realizing their role as universal, ‘off-the-shelf’ treatments requires overcoming translational hurdles. Their theoretical safety and applicability must be further rigorously validated in larger-scale clinical cohorts.

$\gamma\delta$ T cells in hematologic disorders: unmet needs and translational opportunities

Hematologic diseases, whether benign or malignant, represent not only biological disorders but also social and historical challenges, their incidence rising in parallel with aging societies, industrial exposure, and widening healthcare inequality. Even those labeled “non-malignant” conceal severe risks: severe aplastic anemia carries a 1-year mortality of 30–40%, while autoimmune hemolytic anemia reaches 8–23%, with warm autoimmune

hemolytic anemia patients tripling their thromboembolism risk (47–49). Malignancies such as acute myeloid leukemia (AML), myelodysplastic syndromes (MDS), and lymphomas impose an even grimmer reality—relapse is common, survival is poor, and the burden falls heaviest on the elderly or those with hostile genetics (50–52). The five-year survival of elderly AML stays below 20%, while relapsed AML allows only half a year of life (53–56). Access to curative transplantation, limited to <5% in low- and middle-income regions, exposes the structural injustice of our medical systems (9, 57, 58). To this biology we must add the weight of society: exclusion, depression, and financial toxicity, with households spending over half their income on survival, deepening the spiral of loss (59–61).

Matrix of unmet needs in hematologic disorders

Modern therapies, though impressive, falter across several dimensions:

- Efficacy: 40% of severe aplastic anemia patients fail to respond to immunosuppressive therapy (62, 63); 20–30% of diffuse large B-cell lymphoma relapses follow R-CHOP (64–66).
- Toxicity: Bortezomib induces neuropathy in 35–40% of myeloma patients (67–69); ATG drives infection (70, 71); splenectomy triples thrombosis (72, 73).

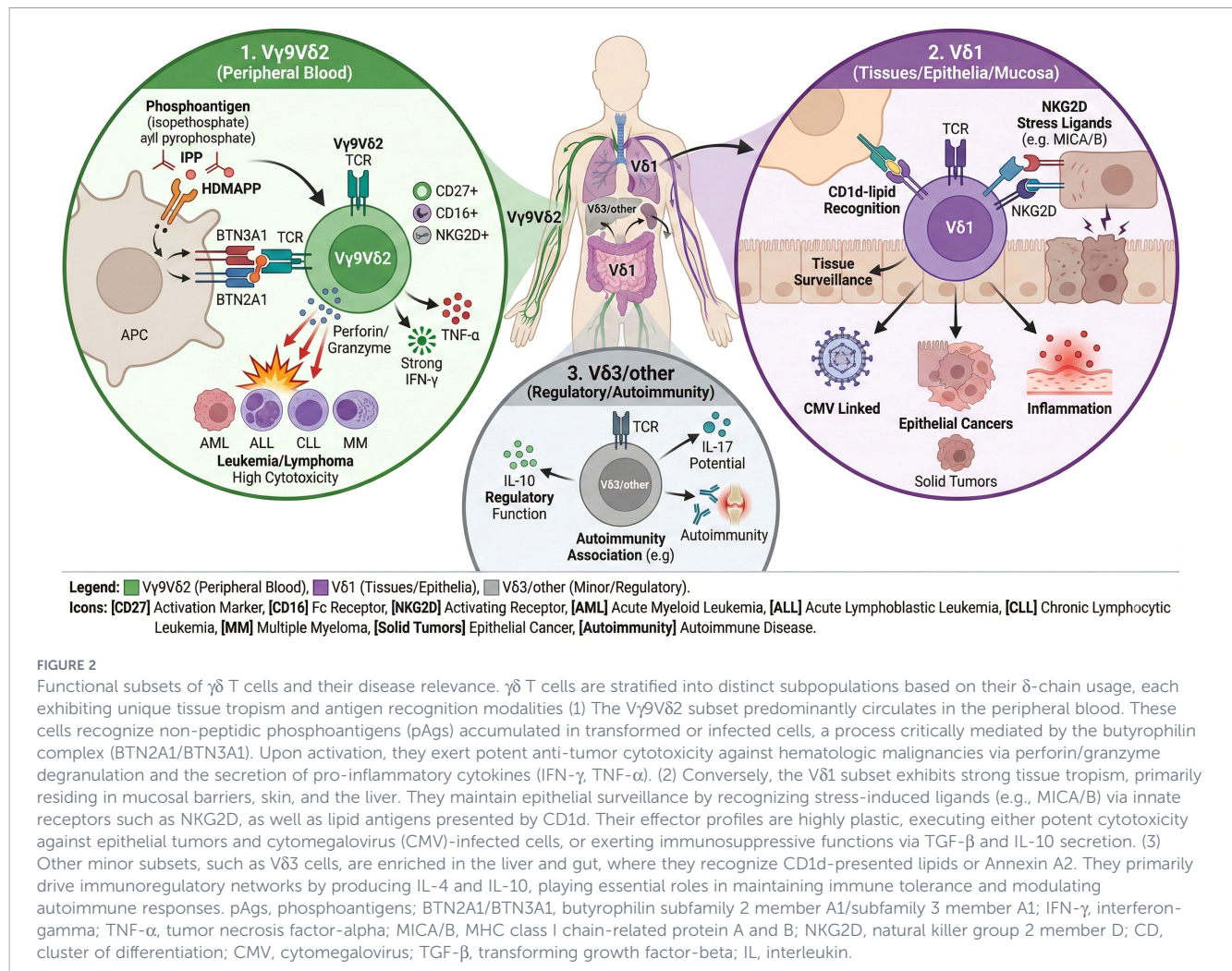


FIGURE 2

Functional subsets of $\gamma\delta$ T cells and their disease relevance. $\gamma\delta$ T cells are stratified into distinct subpopulations based on their δ -chain usage, each exhibiting unique tissue tropism and antigen recognition modalities (1) The Vγ9Vδ2 subset predominantly circulates in the peripheral blood. These cells recognize non-peptidic phosphoantigens (pAgs) accumulated in transformed or infected cells, a process critically mediated by the butyrophilin complex (BTN2A1/BTN3A1). Upon activation, they exert potent anti-tumor cytotoxicity against hematologic malignancies via perforin/granzyme degranulation and the secretion of pro-inflammatory cytokines (IFN-γ, TNF-α). (2) Conversely, the Vδ1 subset exhibits strong tissue tropism, primarily residing in mucosal barriers, skin, and the liver. They maintain epithelial surveillance by recognizing stress-induced ligands (e.g., MICA/B) via innate receptors such as NKG2D, as well as lipid antigens presented by CD1d. Their effector profiles are highly plastic, executing either potent cytotoxicity against epithelial tumors and cytomegalovirus (CMV)-infected cells, or exerting immunosuppressive functions via TGF-β and IL-10 secretion. (3) Other minor subsets, such as Vδ3 cells, are enriched in the liver and gut, where they recognize CD1d-presented lipids or Annexin A2. They primarily drive immunoregulatory networks by producing IL-4 and IL-10, playing essential roles in maintaining immune tolerance and modulating autoimmune responses. pAgs, phosphoantigens; BTN2A1/BTN3A1, butyrophilin subfamily 2 member A1/subfamily 3 member A1; IFN-γ, interferon-gamma; TNF-α, tumor necrosis factor-alpha; MICA/B, MHC class I chain-related protein A and B; NKG2D, natural killer group 2 member D; CD, cluster of differentiation; CMV, cytomegalovirus; TGF-β, transforming growth factor-beta; IL, interleukin.

- c. Accessibility: Merely 35% of nations provide free immunosuppressive therapy (74, 75); transplant-related mortality lingers at 15–25% (76, 77); hematology specialist density spans from 2.3/100,000 in the rich world to 0.04/100,000 in the poor (78).
- d. Immune Evasion: Tumor microenvironments erode T cell vigor (79, 80); HLA loss in AML and MDS transforms therapy into futility (81, 82).

These limitations highlight the need for alternative immune-based therapeutic strategies, and the urgent call for new immune strategies.

Rationale for $\gamma\delta$ T cell-based interventions

Here, $\gamma\delta$ T cells enter as evolutionary outliers with translational promise. Their ability to recognize stress antigens enables direct confrontation with HLA-deficient tumors such as relapsed AML or high-risk MDS (38, 83, 84). Unlike their $\alpha\beta$ cousins, they can enter the marrow, sense cellular stress, and they may exert both cytotoxic and immunoregulatory functions, supporting their translational relevance. (Figure 3).

Summary

Taken together, the biological properties of $\gamma\delta$ T cells suggest clear translational relevance in hematologic diseases. Their MHC-independent recognition, functional plasticity, and low alloreactive potential make them promising candidates for addressing current therapeutic limitations. These features provide the rationale for further exploring their mechanisms of action and clinical applications.

Mechanisms of $\gamma\delta$ T cell activation and function

Multimodal activation pathways enable broad-spectrum immune engagement

Unlike $\alpha\beta$ T cells, $\gamma\delta$ T cells are wide-spectrum detectors, able to detect non-peptidic cues across diverse metabolic and cellular stress pathways (85, 86). (Figure 4). Vγ9Vδ2 T cells in blood respond to phosphoantigens such as isopentenyl pyrophosphate, molecules

TABLE 1 Characteristics and disease associations of $\gamma\delta$ T cell subsets.

Subset	Key receptors/markers	Localization	Cytokine profile & functional plasticity	Associated diseases/roles
V γ 9V δ 2	TCR V γ 9V δ 2, CD27,CD45RA, NKG2D, BTN3A1	Peripheral blood, lymph nodes	Anti-tumor/Pro-inflammatory: IFN- γ , TNF- α (Th1-like). Regulatory: Can be skewed to IL-10 producing state by TME.	Malignancies: Cytotoxicity against AML, ALL, MM via phosphoantigen recognition. Infections: Expansion during viral (CMV, EBV) hematologic injury.
V δ 1	TCR V δ 1, CD1d, NKG2D, NKp30	Mucosal barriers, spleen, liver, bone marrow	Tissue surveillance: IL-17, TGF- β , IL-10. Dual role: Can exhibit potent cytotoxicity (Granzyme/Perforin) or profound immunosuppression.	Malignancies: Elevated in CLL and DLBCL (often exhausted/regulatory). Potent CAR-V δ 1 targets for relapsed AML. Autoimmunity: Dysregulated in ITP and AIHA.
Non-V γ 9V δ 2/V δ 1 (e.g., V δ 3)	TCR V δ 3, CD56, HLA-DR	Liver, gut, rare in blood	Regulatory: IL-10, IL-4, TGF- β . Associated with chronic inflammation and tolerance.	Post-HSCT: Implicated in immune reconstitution and establishing tolerance post-transplant; less defined in direct cytotoxicity.

TCR, T-cell receptor; IFN- γ , interferon-gamma; TNF- α , tumor necrosis factor-alpha; TGF- β , transforming growth factor-beta; IL, interleukin; AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; MM, multiple myeloma; CLL, chronic lymphocytic leukemia; DLBCL, diffuse large B-cell lymphoma; CMV, cytomegalovirus; EBV, Epstein-Barr virus; IBD, inflammatory bowel disease; ITP, immune thrombocytopenia; AIHA, autoimmune hemolytic anemia.

that pile up in tumors where the mevalonate pathway falters (87–89). These phosphoantigens twist BTN3A1 and BTN2A1 into new conformations, unlocking recognition by $\gamma\delta$ TCRs (90–92). Meanwhile, V δ 1+ $\gamma\delta$ T cells register lipid antigens displayed by CD1d-sphingolipids, α -galactosylceramides-eliciting cytokines and repairing barrier tissues (19, 93). Other alarms arise from stress-

induced ligands, MICA, ULBPs, HSP70, activating NKG2D signaling cascades, primarily in V δ 1+ cells, which trigger cytotoxic cascades and inflammatory waves in epithelial sites (94–96). This multi-channel detection system allows $\gamma\delta$ T cells to recognize cancers, infections, and damaged tissue without HLA restriction. Drugs like nitrogen bisphosphonates or CD1d agonists

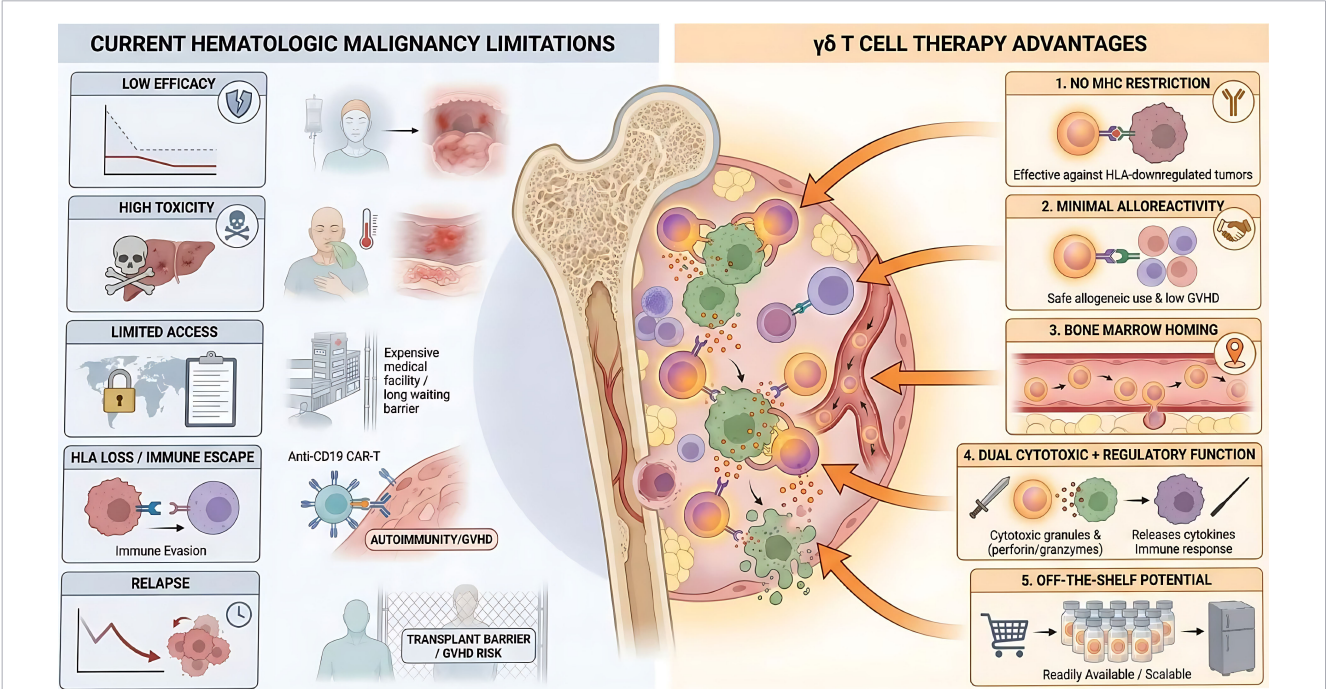


FIGURE 3 Translational advantages of $\gamma\delta$ T cells in hematological diseases. Current conventional immunotherapies face critical clinical bottlenecks, including limited long-term efficacy driven by tumor immune evasion (e.g., HLA downregulation and antigen loss), severe safety toxicities such as cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS), as well as restricted patient accessibility due to high manufacturing costs and prolonged vein-to-vein times. In contrast, $\gamma\delta$ T cells offer intrinsic translational solutions to these barriers (1) Their MHC-independent recognition enables the direct targeting and eradication of HLA-deficient tumor variants that evade conventional $\alpha\beta$ T cell surveillance. (2) Their unique biology confers a highly favorable safety profile, characterized by a minimal risk of graft-versus-host disease (GVHD) in allogeneic settings and a significantly lower incidence of severe CRS or ICANS. (3) Intrinsic chemokine receptor expression facilitates robust tissue homing and deep infiltration into protective tumor niches, such as the bone marrow. (4) Finally, their dual functionality allows them to execute direct potent cytotoxicity against malignant blasts while simultaneously orchestrating broader adaptive immune responses within the immunosuppressive tumor microenvironment (TME). HLA, human leukocyte antigen; CRS, cytokine release syndrome; ICANS, immune effector cell-associated neurotoxicity syndrome; MHC, major histocompatibility complex; GVHD, graft-versus-host disease; TME, tumor microenvironment.

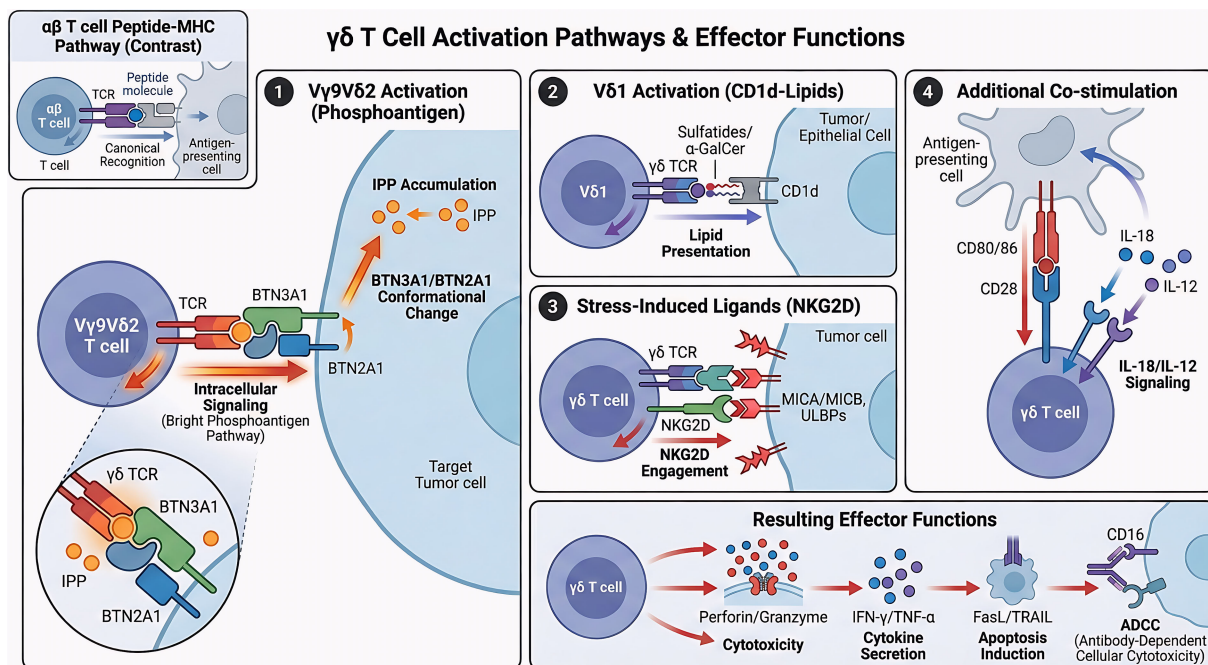


FIGURE 4

Multimodal activation mechanisms of $\gamma\delta$ T cells. Conventional $\alpha\beta$ T cells strictly depend on the recognition of specific peptide antigens presented by major histocompatibility complex (MHC) molecules on antigen-presenting cells (APCs). Conversely, $\gamma\delta$ T cells utilize a broad, MHC-independent, and multimodal recognition system to detect transformed or infected cells. (1) Phosphoantigen Recognition: The $\gamma\delta$ T-cell receptor (TCR) senses intracellular metabolic dysregulation by detecting accumulated phosphoantigens (pAgs, such as IPP or microbial HMBPP), a structural engagement mediated by the butyrophilin complex (BTN2A1 and BTN3A1). (2) Lipid Recognition: The $\gamma\delta$ TCR independently recognizes lipid and glycolipid antigens presented by the non-classical MHC-like molecule CD1d. (3) Innate-like Stress Recognition: Synergizing with TCR signals, innate activating receptors such as NKG2D directly engage stress-induced surface ligands (e.g., MICA and MICB) that are upregulated on malignant cells. These converging activation pathways culminate in robust downstream effector functions, driving direct tumor cytotoxicity (via perforin and granzyme B degranulation) and the secretion of pro-inflammatory cytokines (IFN- γ and TNF- α) to orchestrate broader immune responses. APC, antigen-presenting cell; MHC, major histocompatibility complex; TCR, T-cell receptor; pAgs, phosphoantigens; IPP, isopentenyl pyrophosphate; HMBPP, (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate; BTN2A1/3A1, butyrophilin subfamily 2 member A1/subfamily 3 member A1; CD1d, cluster of differentiation 1d; NKG2D, natural killer group 2 member D; MICA/B, MHC class I chain-related protein A and B; IFN- γ , interferon-gamma; TNF- α , tumor necrosis factor-alpha.

can tune these activation channels, selectively expanding tumor-reactive $\gamma\delta$ T cells in controlled fashion, providing scalable therapeutic levers (97–99).

Cytokine plasticity as a determinant of functional programming and disease stratification

$\gamma\delta$ T cells exhibit high functional plasticity, which is heavily modulated by subset identity and the surrounding field. V γ 9V δ 2 cells usually polarize into Th1-like states, secreting IFN- γ and TNF- α via NF- κ B and MAPK, amplifying tumor immunity and macrophage action (100–102). Yet in tumor microenvironments steeped in TGF- β and IL-6, $\gamma\delta$ T cells may undergo phenotypic skewing into IL-10 or IL-4 producers, suppressing localized immune responses (103–105). V γ 6V δ 1 subsets in mucosa can be prolific IL-17A/F sources-bolstering barrier defense or, conversely, fueling chronic inflammation in autoimmunity and myeloid cancers (106–108).

Decoding this cytokine plasticity allows precision use of $\gamma\delta$ T cells. IFN- γ -dominant subsets suit tumor therapy (109–111), while IL-10-producing $\gamma\delta$ Tregs may drive resistance or be harnessed against inflammation (112, 113). Tuning these cytokine circuits offers both stratification and engineering routes for therapy.

Dual cytotoxic and immune reprogramming functions drive antitumor efficacy

$\gamma\delta$ T cells can precisely target lesions and also modulate the local microenvironment post-cytolysis. They are involved in mechanisms such as perforin/granzyme granules, FasL/TRAIL apoptosis, antibody-dependent cell-mediated cytotoxicity (ADCC), effectively eliminating malignant or infected cells (114, 115). V γ 9V δ 2 cells sense metabolic distress (116, 117), while V δ 1 cells engage stress ligands in tissues (118, 119). Beyond killing, $\gamma\delta$ T cells rewire the tumor microenvironment: releasing chemokines such as CCL3, CXCL8, and TH1 cytokines that call in dendritic cells, NK cells, CD8+ T cells, creating long-term immune reinforcement (120, 121).

The dual characteristics of these cytotoxic effectors and immunomodulators make $\gamma\delta$ T cells prime for immunotherapy. Adoptive transfers, including CAR- $\gamma\delta$ T cells, exploit this versatility; early-phase human clinical trials and preclinical *in vivo* animal models show tumor suppression through both direct cytotoxicity and microenvironmental immune reprogramming (122–124). Pairing $\gamma\delta$ therapy with checkpoint inhibitors or metabolic adjuvants may unlock resistant disease.

Cooperative crosstalk with $\alpha\beta$ T cells, NK cells, and dendritic cells underpins combination therapy

$\gamma\delta$ T cells function as the connectors, bridging immune constellations. Through IFN- γ and GM-CSF they energize dendritic cells to sharpen antigen presentation (125–127); through 4-1BBL and chemokines they elevate NK cytotoxicity (128, 129); through chemotactic cues they draw in CD8⁺ T cells (130–132). Under inflammation, these synergies amplify, and $\gamma\delta$ T cells also present antigens themselves or secrete AREG to mend tissues (133, 134).

Harnessing $\gamma\delta$ T cells as adjuvants can amplify other therapies— $\alpha\beta$ T cells, oncolytic viruses, dendritic cell vaccines. Rational co-culture systems and multi-cell adoptive designs are the emerging blueprints for such synergy.

Functional plasticity as the basis for $\gamma\delta$ T cell versatility in clinical applications

The versatility of $\gamma\delta$ T cells lies in their ability to detect varied antigens, secrete a broad cytokine palette, combine cytotoxic with regulatory roles, and weave networks with other immune subsets (135–137). Their independence from MHC, their sensitivity to metabolism, and their effector-regulatory duality situate them at the crossroads of cancer, infection, and immune imbalance (138–140).

Functional plasticity is the key factor of $\gamma\delta$ T cell therapy. Engineering expansion and modulation to favor IFN- γ + cytotoxic or IL-10+ regulatory states offers precise tailoring (141–143). As clinical trials unfold, predictive biomarkers and refined manufacturing pipelines will determine how far this versatile immune axis can be pushed into practice.

$\gamma\delta$ T cells in hematologic malignancies: translational opportunities and challenges

Critical limitations: clinical discrepancies, durability, and expansion variability

Even with remarkable therapeutic advances, hematologic malignancies—acute leukemia, lymphoma, and multiple myeloma (MM)—remain formidable. They are governed by invisible forces of relapse, resistance, and immune escape (144–146). In relapsed or refractory acute lymphoblastic leukemia (ALL), elderly patients face a five-year event-free survival below 30% (147–149). Acute myeloid leukemia (AML) offers no reprieve: in older cohorts, fewer than one in five reach the five-year mark, and relapse shortens life to 6–8 months (150, 151). Chronic lymphocytic leukemia (CLL), when burdened with TP53 mutations, yields less than 30% five-year survival, with resistance to BTK inhibitors surpassing 60% (152–154). In MM, patients carrying del(17p), t(4, 14), or TP53 mutations confront swift relapse (155). Lymphomas, from DLBCL to Hodgkin's disease, present persistent clinical challenges,

including immune evasion, CAR-T resistance, recurrence (156–158). Transplant relapse and minimal residual disease (MRD) negatively impact long-term patient survival and treatment durability, underscoring the need for innovative $\gamma\delta$ T-cell immunotherapies (159–161).

Mechanisms of action: subset-specific immune surveillance

$\gamma\delta$ T cells connect innate and adaptive immunity to execute subset-specific surveillance. V γ 9V δ 2 cells perceive phosphoantigens such as IPP, deploying NKG2D to sense MICA/B, and strike with IFN- γ and granzyme B (162–164). V δ 1 cells, guardians of hidden niches, target stress ligands like CD123, infiltrating marrow sanctuaries and eradicating leukemic stem cells (LSCs) (165–167). In preclinical *in vitro* and murine xenograft models of AML and ALL, they have been shown to dismantle CD34+CD19+ reservoirs of relapse (168–170). Yet in CLL and MM, V δ 1 can be bent toward suppression, secreting IL-10 and IL-13 (171–173). Meanwhile, V γ 9V δ 2 cells are frequently inhibited by metabolically restrictive and checkpoint-upregulated microenvironments, PD-1, TIM-3, and mitochondrial collapse impairing their cytotoxic efficacy (174–176). These subset-specific dynamics mirror species diversity, and guide the design of targeted therapies.

Clinical and preclinical evidence supporting $\gamma\delta$ T therapies

Distinguishing preclinical efficacy from early-phase clinical outcomes

Preclinical models evaluating $\gamma\delta$ T-cell therapies have consistently demonstrated anti-leukemic activity; however, translating these preclinical milestones into robust and predictable clinical outcomes remains a challenge. It requires a rigorous distinction between preclinical models and human clinical trials (177). In early-phase human clinical trials for ALL, expanded V γ 9V δ 2 reinfusion has achieved complete remission (CR) in approximately 38% of small patient cohorts, while early clinical evaluations of CD19-CAR- $\gamma\delta$ T cells report response rates up to 67% (178–180). In human cohorts treated with decitabine combinations, the overall response rate (ORR) rose to 45% (181).

Conversely, strictly within preclinical murine xenograft models, experimental therapies such as NKG2D-CAR- $\gamma\delta$ T cells and anti-CD123 CAR-V δ 1 cells have yielded profound leukemia clearance rates of 52% and 63% in AML, respectively. Similarly, in preclinical *in vivo* models of high-risk CLL, ROR1-CAR-V δ 1 and PD-1-CAR- $\gamma\delta$ combinations have demonstrated potent tumor suppression (CR equivalent to 65% and ORR to 54% in murine cohorts), awaiting human translation (182–184).

Returning to human clinical data, Phase I trials involving patients with multiple myeloma (MM) infused with anti-BCMA CAR-V γ 9V δ 2 have achieved encouraging results, with a CR of 65% and an 82% overall survival (OS) rate at 24 months in small cohorts (185–187). In human Phase I dose-escalation trials for relapsed/refractory B-cell lymphoma, the allogeneic CD20-targeted CAR-V δ 1 therapy (ADI-001) has delivered an ORR of 62% and a CR of

41% without severe toxicities (188–190). Finally, retrospective correlative observations in human transplant recipients show that robust V δ 1/V δ 2 recovery significantly correlates with MRD negativity (91%) and a reduction in relapse risk (HR = 0.32) (189, 191–193).

Current human clinical data are predominantly restricted to Phase I/II safety and dose-escalation trials with small patient cohorts (typically N < 20). These early-phase clinical observations confirm the exceptional safety profile of allogeneic $\gamma\delta$ T cell infusions (with negligible severe GVHD), but they also reveal that the sustained clinical efficacy and *in vivo* persistence often fall short of the dramatic responses observed in animal models. To provide clear delineation, the following sections will explicitly categorize the evidence levels, and a comprehensive summary of recent patient-derived clinical trial data is synthesized in Table 2.

Epigenetic and transcriptional regulation of $\gamma\delta$ T cells by the hematologic tumor microenvironment

The hematologic tumor microenvironment (TME) exerts profound modulatory effects on $\gamma\delta$ T cell polarization, often driving an immunosuppressive phenotype through complex epigenetic and transcriptional reprogramming. Recent advances, including spatial transcriptomic analyses, have unveiled highly compartmentalized niches that dictate immune evasion (194).

A prominent example is observed within the diffuse large B-cell lymphoma (DLBCL) microenvironment. Spatial mapping reveals that tumor-derived signals heavily influence the local $\gamma\delta$ T cell state (195). Transcriptional networks driven by the TGF- β pathway—specifically through the TGF β 1-IL2RA signaling axis—can skew $\gamma\delta$ T cells toward an IL-10-producing, regulatory phenotype ($\gamma\delta$ Tregs), effectively suppressing localized cytotoxic responses and promoting lymphoma progression.

Furthermore, epigenetic remodeling within the hematologic TME upregulates inhibitory checkpoints. The LGALS9-CD44 axis

has recently emerged as a critical immune checkpoint regulator in this context (196). Galectin-9 (LGALS9), secreted by malignant cells or associated stroma, engages CD44 on infiltrating $\gamma\delta$ T cells. This interaction initiates downstream transcriptional programs that drive functional exhaustion, metabolic suppression, and premature apoptosis. Conversely, in acute myeloid leukemia (AML), the hypoxic bone marrow niche drives epigenetic silencing of NKG2D expression via histone deacetylase (HDAC) activity, further paralyzing V δ 1 subset surveillance (197–199). Understanding these specific spatial and transcriptional networks is crucial for reversing TME-induced exhaustion in $\gamma\delta$ T cell therapies.

Translational barriers and strategies

Critical limitations: clinical discrepancies, durability, and expansion variability

Metabolic suppression-IDO consuming tryptophan, arginase draining arginine-further darkens their energy (200, 201). Suppressive ligands like Galectin-9 and IL-13 further exacerbate localized immunosuppression (202, 203). Yet pathways to escape exist: checkpoint blockade (anti-PD-1, TIM-3, LAG-3), metabolic interventions (epacadostat, CB-839), and cytokine fuel (IL-2, IL-15) (204–206). Engineering expands their reach-dual CARs (CD19/CD20), IL-15R, bispecific engagers (207–209). Combinations-cytarabine with $\gamma\delta$ T cells (AML, overall response rate: 58%), pomalidomide with V δ 1 inhibition (MM, overall response rate: 45%), CD19/CD20 CARs in diffuse large B-cell lymphoma (relapse to 15%)-illuminate synergistic effects (210–212). Graft engineering, purging $\alpha\beta$ TCRs, lessens GVHD while preserving GVL, bending the arc of immunity (213–215).

Rational engineering and CAR design considerations for $\gamma\delta$ T cells

To overcome the hostile hematologic TME, rational engineering of Chimeric Antigen Receptors (CARs) specific to $\gamma\delta$ T cells is

TABLE 2 Recent key clinical trials of $\gamma\delta$ T cell therapies in hematologic malignancies.

Disease	Intervention/subset	Clinical trial phase	Cohort/N	Main outcomes & efficacy	Safety profile/adverse events	Ref
Relapsed/Refractory B-NHL	LUCAR-G39D (Allogeneic Anti-CD20/CD19 Dual-CAR $\gamma\delta$ T cells)	Phase I (NCT06395870)	N=12	88% of evaluable patients showed decreased ctDNA; encouraging early ORR and durability.	Manageable safety. Infections observed (mostly Grade 1/2), no severe ICANS.	<i>Blood</i> (2025) 146:552698
High-Risk AML (Post-alloSCT)	Ex vivo expanded donor-derived allogeneic V γ 9V δ 2 T cells	Phase I (NCT05015426)	N=12	17% relapse rate at 11 months median follow-up. Majority remained in CR.	Highly favorable. No dose-limiting toxicities or severe acute GVHD induced.	<i>VJHemOnc</i> (2025); <i>Front Immunol</i>
B Cell Malignancies (MCL focus)	ADI-001 (Allogeneic CD20-CAR V δ 1 T cells)	Phase I (NCT04735471)	N=10 (MCL cohort)	ORR 80%, CR 60%; mDoR 17.5 months. Highly competitive with approved agents.	No Grade $\geq$ 3 CRS or ICANS. No GVHD reported.	<i>ASCO/Adicet Data</i> (2024-2025)
CD123+ R/R AML & MDS	LAVA-1266 (CD123-targeting bispecific V γ 9V δ 2-T cell engager)	Phase I (ACTRN12624001214527)	Escalation cohort (~50 planned)	Dose escalation ongoing. Preclinical validation showed potent CD123+ blast clearance.	Interim safety shows limited off-target toxicity and manageable cytokine profiles.	<i>WSLHD Repository</i> (2025)

R/R, relapsed/refractory; B-NHL, B-cell non-Hodgkin lymphoma; alloSCT, allogeneic stem cell transplantation; MCL, mantle cell lymphoma; MDS, myelodysplastic syndromes; ORR, overall response rate; CR, complete remission; PR, partial response; mDoR, median duration of response; ctDNA, circulating tumor DNA; CRS, cytokine release syndrome; ICANS, immune effector cell-associated neurotoxicity syndrome; GVHD, graft-versus-host disease.

imperative. Unlike conventional $\alpha\beta$ T cells, $\gamma\delta$ T cells benefit from unique co-stimulatory requirements (216).

Optimized Co-stimulatory Domains: Standard CD28 or 4-1BB endodomains may not fully capitalize on $\gamma\delta$ T cell biology. Incorporating NKG2D or CD27 intracellular signaling domains into the CAR construct has been shown to synergize with innate TCR signaling, reducing activation-induced cell death (AICD) and enhancing long-term persistence (217).

“Armored” CAR Constructs: To combat transcriptional exhaustion *in vivo*, engineering $\gamma\delta$ T cells to constitutively express supportive cytokines (e.g., tethered IL-15 or IL-18) maintains metabolic fitness without the need for systemic cytokine administration (218).

Logic-Gated Systems: To mitigate on-target, off-tumor toxicity in myeloid malignancies (such as targeting CD123 or CLL-1), AND-gated or OR-gated CAR systems are being evaluated. These require the simultaneous engagement of a synthetic tumor antigen and an innate stress ligand (via native NKG2D), ensuring highly localized cytotoxicity exclusively within the leukemic marrow niche (219).

Research recommendations and priority areas

There are five priority areas as follows:

- Stratify patients by V δ 1/V δ 2 balance and checkpoint expression;
- Improve *in vitro* expansion with zoledronic acid or IL-15;
- Create disease-specific CAR- $\gamma\delta$ constructs—CD123 for AML, ROR1 for CLL, BCMA for MM;
- Pair metabolic reprogramming with checkpoint inhibition;
- Advance phase II/III trials in minimal residual disease-positive and transplant-relapsed cohorts.

The evidence level is strongest in AML, ALL, and MM-marked “High” priority. CLL and lymphomas remain less charted, yet with engineering and microenvironmental mastery, they too may yield.

$\gamma\delta$ T cells in hematologic disorders beyond malignancies

Immune tolerance reconstruction in autoimmune hematologic disorders (AIHA, ITP, AA)

Autoimmune hematologic disorders, such as autoimmune hemolytic anemia, immune thrombocytopenia, and aplastic anemia, are the outcome of immune system attacking autologous blood cells. Past methods—glucocorticoids, rituximab, immunosuppressants—seek to suppress, but not to solve (220–222). They transiently suppress inflammation but rarely restore long-term immune tolerance. $\gamma\delta$ T cells, with their unique and flexible role, step forward as modulators of dysregulated immune responses. In hemolytic anemia,

experiments in mice show that activated $\gamma\delta$ T cells drive B-1 cells, which in turn exacerbate autoantibody-mediated destruction (223–225). But under clear conditions, the same $\gamma\delta$ T cells, secreting IL-10 and TGF- β , can suppress autoreactive B cell clones, and keep immune tolerance (226–228). In ITP, $\gamma\delta$ T cells bear two faces: some hasten platelet loss, while others strengthen thrombopoietin receptor signals, aiding Tregs in restraining autoimmunity (229–231). In aplastic anemia, they may stand guard over stem cells, or they may strike against them, depending on the camp they take (232–234). Thus in the field of immunotherapy, a viable strategy has been identified: to select the appropriate subset of immune cells, design suitable stimuli to activate these cells, enrich the V δ 1+ cells that produce IL-10, and place them at the forefront of immune reconstruction. This process points to a potential approach that could achieve long-term disease remission and prevent recurrence.

$\gamma\delta$ T cells as adjunctive agents in infection-induced hematologic injury

Factors that activate the immune system are not limited to autoimmune diseases; sometimes they appear in the form of infections—such as HIV, tuberculosis, and malaria—which attack the blood and bone marrow, disrupting the human immune system (235–237). Conventional therapy clears microbes, but cannot support the restoration of the hematopoietic niche (238–240). Here again, $\gamma\delta$ T cells have their duty. In tuberculosis, V γ 9V δ 2 T cells sense phosphoantigens, call BTN3A1 into play, and release IFN- γ and granulysin, killing the invader (241–243). In HIV infection, the subset distribution shifts: the V δ 2+ population contracts while the V δ 1+ population expands, disrupting the CD4+/CD8+ balance (244–246). Yet in those exposed but uninfected, $\gamma\delta$ T cells secrete chemokines that resist the virus-proof of their protective power (247, 248). In malaria, $\gamma\delta$ T cells rally Th1-type defenses and hold the line against parasitemia, though prolonged struggle wears them down (249–251). What is to be done? Restore their vigor, engineer their strength, precondition with cytokines, combine them with marrow-supporting agents—thus one may both strike the pathogen and preserve the niche. This is a new road in translational strategy.

Subset optimization and preconditioning for functional specificity

$\gamma\delta$ T cells are not a single entity but are diverse. Some cells secrete IFN- γ and TNF- α to activate immune cells, while others produce IL-10 and TGF- β to suppress immune activation (252–254). V γ 9V δ 2 cells are renowned for cytotoxicity and defense against infection (255–257). V δ 1+ cells, with tissue-homing and tolerogenic gifts, play another part (258). Preclinical work shows: by pre-treating immune cells with specific cytokines (IL-2, IL-15, IL-21), their effector capabilities can be enhanced or strengthened (259–261). On the other hand, when these cells are exposed to inhibitory cytokines such as IL-10 or TGF- β , they shift towards a more suppressive or regulatory role, which helps maintain the balance of the immune system, prevent overreaction, and reduce inflammation (18, 262, 263). Sorting by CD27, CD45RA, and

shaping with epigenetic tools brings fidelity and order. For the clinic, precision must match condition: V δ 2 cells rich in IFN- γ for clearing infection, V δ 1 cells rich in IL-10 for rebalancing autoimmunity (264–267). Thus, by tailoring expansion, we avoid waste, reduce danger, and secure maximum benefit in hematologic disease.

Translational priorities and clinical development directions

Though evidence grows, $\gamma\delta$ T cells are still rare in clinical trials of blood disease. This is a shortcoming, and must be corrected. Three priorities lie before us: first, use IL-10+ $\gamma\delta$ T cells to induce tolerance in stubborn autoimmune disorders; second, deploy $\gamma\delta$ T cell transfer as an ally against drug-resistant tuberculosis; third, build subset-specific platforms with safety switches, to command and recall the cells at will. Trials must not be blind—they must track subsets by biomarkers, profile them at single-cell level, and measure outcomes not only by response, but by reconstitution, relapse, and side effects. To make the therapy accessible, prepare off-the-shelf products from healthy donors, expand them feeder-free, and secure GMP standards. In conclusion, while $\gamma\delta$ T cells represent a promising vector for cellular immunotherapy, they are currently in the developmental and optimization stage. Addressing the critical limitations of *in vivo* persistence, functional exhaustion, and manufacturing consistency is paramount for their future clinical integration.

Therapeutic potential of targeting $\gamma\delta$ T cells

Overview of $\gamma\delta$ T cell–based therapeutic platforms

The field of $\gamma\delta$ T cell therapy has opened up many paths (117, 268, 269) (Figure 5). We can use adoptive cell transfer, we can build chimeric antigen receptors (CAR), we can design bispecific antibodies, and we can apply pharmacologic activators like bisphosphonates. We can also combine these approaches. The strength of $\gamma\delta$ T cells across these platforms lies in their natural movement to tissues and their swift effector function (270–272) (Table 3). These are advantages that, if fully used, can make them important in cancer immunotherapy.

Translational bottlenecks and proposed solutions

Despite the compelling rationale for $\gamma\delta$ T cell-based therapies, a critical appraisal of current literature reveals substantial limitations that must temper clinical expectations. Addressing these limitations will require improvements in expansion protocols, product consistency, and *in vivo* persistence. The answer lies in improving methods of expansion, reducing variability, and finding ways to strengthen persistence (Figure 6). Only then can we turn promise into real effect.

First, there is a marked discrepancy between preclinical success and clinical efficacy. While *in vitro* assays and murine xenograft models consistently show robust cytolytic clearance of hematologic blasts by expanded $\gamma\delta$ T cells, these models fail to fully recapitulate the complex, immunosuppressive human tumor microenvironment (TME) (273). Human phase I trials frequently report modest overall response rates, highlighting that preclinical cytotoxicity does not seamlessly translate to *in vivo* tumor eradication, largely due to the human TME's capacity to induce rapid anergy (274).

Second, the durability of clinical responses remains a significant limitation. Unlike CD19-targeted $\alpha\beta$ CAR-T cells, which can exhibit prolonged *in vivo* expansion and memory formation, adoptively transferred $\gamma\delta$ T cells often suffer from rapid clearance and a lack of long-term persistence without continuous exogenous cytokine support (e.g., IL-2 or IL-15) (275). *In vivo*, these cells rapidly upregulate exhaustion markers such as PD-1 and TIM-3 upon engaging the leukemic niche, drastically curtailing their sustained effector function and leading to transient responses.

Finally, manufacturing variability remains a profound translational bottleneck. The *ex vivo* expansion of $\gamma\delta$ T cells (primarily the V γ 9V δ 2 subset) using aminobisphosphonates is heavily donor-dependent. High variability in final cell yield, purity, and functional fitness is frequently observed, particularly when utilizing autologous cells from heavily pre-treated patients whose baseline immune compartments are already compromised (276). This donor-to-donor heterogeneity severely challenges the development of standardized, reproducible 'off-the-shelf' allogeneic products, underscoring the necessity for optimized, feeder-free GMP manufacturing protocols before widespread clinical adoption can be realized.

Challenges and future directions in $\gamma\delta$ T cell therapy

Challenges in $\gamma\delta$ T cell product consistency

In the short period of two years, it will be necessary to set the standards of control and to define markers of function. Over five years, practice must converge toward protocols that are uniform and suitable for GMP facilities in many centers.

Unresolved biology and targeting of $\gamma\delta$ TCRs

The next question is biological in nature. Although experiments in the laboratory suggest potential, the full understanding of $\gamma\delta$ T cells has not been attained. One observes that no single transcription factor defines a subtype, so knockout mice cannot be easily constructed. V δ 1 and V γ 9V δ 2 differ, but the precise criteria for choosing one or the other in patients remain uncertain (277). Their migration in tissues, their response to gradients of phosphoantigen, and their ability to enter tumors are all incompletely known. For progress, new tools are required: organoid models, single-cell technologies, and gene-engineered animals such as TCR γ -KO

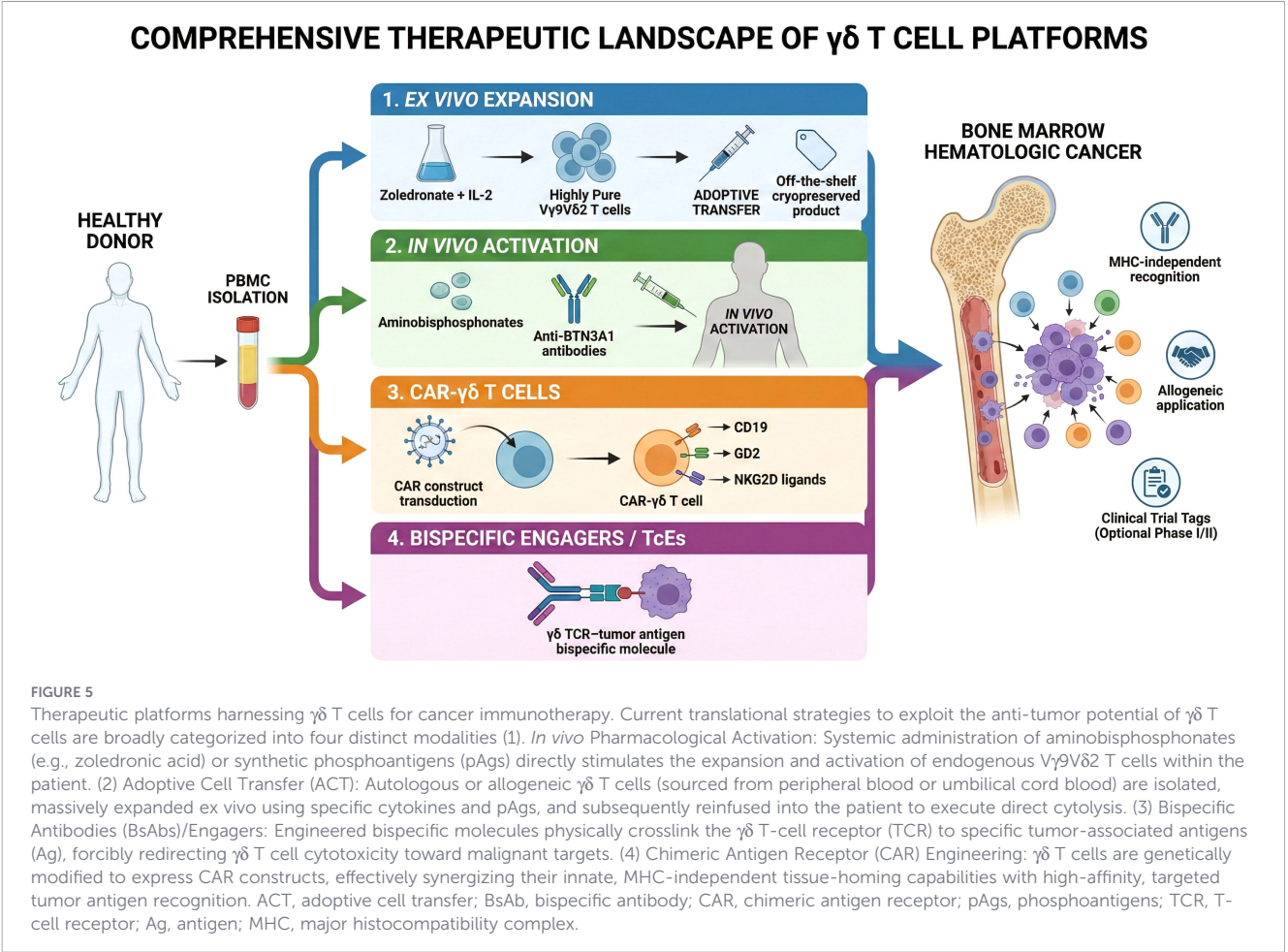


FIGURE 5
Therapeutic platforms harnessing $\gamma\delta$ T cells for cancer immunotherapy. Current translational strategies to exploit the anti-tumor potential of $\gamma\delta$ T cells are broadly categorized into four distinct modalities (1). *In vivo* Pharmacological Activation: Systemic administration of aminobisphosphonates (e.g., zoledronic acid) or synthetic phosphoantigens (pAgs) directly stimulates the expansion and activation of endogenous V γ 9V δ 2 T cells within the patient. (2) Adoptive Cell Transfer (ACT): Autologous or allogeneic $\gamma\delta$ T cells (sourced from peripheral blood or umbilical cord blood) are isolated, massively expanded ex vivo using specific cytokines and pAgs, and subsequently reinfused into the patient to execute direct cytotoxicity. (3) Bispecific Antibodies (BsAbs)/Engagers: Engineered bispecific molecules physically crosslink the $\gamma\delta$ T-cell receptor (TCR) to specific tumor-associated antigens (Ag), forcibly redirecting $\gamma\delta$ T cell cytotoxicity toward malignant targets. (4) Chimeric Antigen Receptor (CAR) Engineering: $\gamma\delta$ T cells are genetically modified to express CAR constructs, effectively synergizing their innate, MHC-independent tissue-homing capabilities with high-affinity, targeted tumor antigen recognition. ACT, adoptive cell transfer; BsAb, bispecific antibody; CAR, chimeric antigen receptor; pAgs, phosphoantigens; TCR, T-cell receptor; Ag, antigen; MHC, major histocompatibility complex.

mice. The task of the coming five years is to map these interactions in a systematic way, and to obtain a coherent picture of the $\gamma\delta$ TCR in relation to tumor ligands.

Clinical standardization and biomarker gaps

Turning to the clinic, one again meets the problem of divergence. Each center defines subpopulations in its own manner; flow cytometry gates are not alike (278). Some investigators emphasize function, others emphasize purity. This makes results difficult to compare. Yet there are first indications of common measures: levels of HMBPP, receptors such as NKG2D or CD16A, and metabolic

markers like SDHA (279–281). It follows that the near goal-within two years-is a shared immunophenotyping panel and infusion method. At five years, the requirement will be greater: biomarkers must guide which patients should receive therapy, and pharmacodynamic measures must enter into trial design. Only thus can results be both reproducible and useful.

Strategic priorities and translational timeline

From these facts arises the need for order in research. The immediate task is practical: $\gamma\delta$ T cells must be expanded with little exhaustion and tested for survival after infusion. In the next five

TABLE 3 $\gamma\delta$ T cell therapeutic platform overview.

Platform	Example product/modality	Target	Indication	Clinical status
ACT	Expanded V γ 9V δ 2 T cells	–	AML, glioblastoma	Phase I–II
CAR- $\gamma\delta$ T	ADI-001 (CD20-CAR-V δ 1), CD19 CAR-V δ 2	CD20, CD19	B-cell lymphoma	Phase I (Fast Track)
Bispecific $\gamma\delta$ T engagers	BsAb (e.g., CD3 $\times$ $\gamma\delta$ TCR)	Tumor Ag/ $\gamma\delta$ TCR	Solid tumors	Preclinical–Early Trial
Pharmacologic activators	Zoledronic acid + IL-2	FPPS/BTN3A1	Breast cancer, AML, post-HSCT	Pilot trials
Combinatorial	$\gamma\delta$ T + ICI/radiotherapy/chemotherapy	–	Melanoma, lung cancer	Exploratory Phase

ACT, adoptive cell transfer; CAR, chimeric antigen receptor; BsAb, bispecific antibody; Ag, antigen; FPPS, farnesyl pyrophosphate synthase; BTN3A1, butyrophilin subfamily 3 member A1; ICI, immune checkpoint inhibitor; AML, acute myeloid leukemia; HSCT, hematopoietic stem cell transplantation.

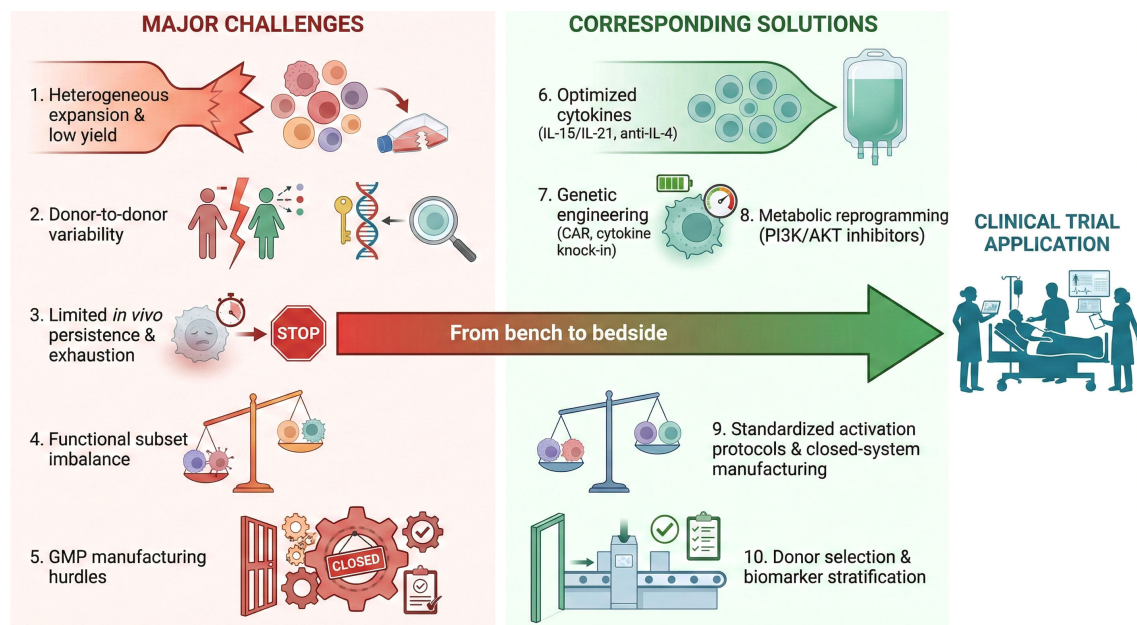


FIGURE 6

Translational bottlenecks and solutions for $\gamma\delta$ T cell therapy. The successful transition of $\gamma\delta$ T cell therapies from bench to bedside is currently hindered by several critical challenges. These include (1) heterogeneous ex vivo expansion resulting in low cell yields, (2) significant donor-to-donor intrinsic variability, (3) limited *in vivo* persistence driven by rapid functional exhaustion, (4) imbalances among functional subsets (e.g., cytotoxic versus regulatory phenotypes), and (5) stringent Good Manufacturing Practice (GMP) production hurdles. To overcome these barriers, corresponding multi-dimensional solutions are actively deployed: (6) utilizing optimized cytokine combinations (e.g., IL-15, IL-21, and anti-IL-4 antibodies) to selectively drive the proliferation of effector subsets; (7) employing genetic engineering (such as CAR constructs or cytokine knock-ins) to enhance autonomous survival; (8) applying metabolic reprogramming strategies (e.g., PI3K/AKT pathway inhibitors) to prevent terminal exhaustion; (9) establishing standardized activation protocols within closed-system manufacturing to ensure product consistency; and (10) implementing precise donor selection and biomarker stratification. The integration of these strategies is essential to optimize the efficacy, reproducibility, and scalability of clinical trial applications. GMP, Good Manufacturing Practice; IL, interleukin; CAR, chimeric antigen receptor; PI3K, phosphoinositide 3-kinase; AKT, protein kinase B.

years, more advanced products-CAR-modified $\gamma\delta$ T cells and bispecific adaptors-should be tested in both solid and blood cancers. The longer view, of about eight years, is that one should aim for cells prepared in advance, edited for resistance to chemotherapy, and capable of durable tumor control. This vision depends on three further steps: reliable biomarkers to predict effect, careful study of dose, and means to overcome suppression by the tumor itself.

Building collaborative platforms and data sharing

Finally, such work cannot be done by single groups in isolation. One must imagine a consortium of centers, in which patients are enrolled by uniform rules, cells are prepared by common methods, and data are analyzed with shared tools. Bioinformatics will integrate multi-omics results with clinical outcomes, and machine learning will help to find predictive signs (282–284). Within two years, agreements for sharing data and establishing registries must be put in place. In five years, one should expect multi-center trials, employing harmonized products and common endpoints. Only by such collaboration can the path to regulatory approval and practical treatment of patients be made smooth.

Safety profiles and risk management in $\gamma\delta$ T cell therapy

The unique safety profile and risk considerations of $\gamma\delta$ T cells must be rigorously identified and managed.

First, Immune-Mediated Toxicities: Although severe Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS) are significantly less frequent compared to conventional autologous $\alpha\beta$ CAR-T therapies, transient grade 1–2 CRS is frequently observed, particularly upon *in vivo* activation (e.g., using aminobisphosphonates) or following the infusion of highly active CAR- $\gamma\delta$ T cells (285).

Second, Pathological Plasticity and Unintended Regulatory Effects: The extreme inherent plasticity of $\gamma\delta$ T cells poses a theoretical risk of pathological transdifferentiation within the tumor microenvironment (TME). Under conditions of chronic inflammation or specific cytokine milieus (such as TGF- β rich niches), effector subsets may skew towards an unintended regulatory phenotype ($\gamma\delta$ Tregs) or an IL-17-producing ($\gamma\delta$ T17) state. These subsets can inadvertently suppress localized anti-tumor immunity, promote angiogenesis, and potentially exacerbate co-existing autoimmune conditions (286).

Furthermore, manufacturing variability remains a critical risk factor for clinical translation. The intrinsic heterogeneity of donor starting material leads to inconsistent *ex vivo* expansion rates and variable product purity, directly impacting dose reproducibility. Establishing standardized quality control markers and robust, reproducible GMP protocols is a paramount priority for comprehensive risk mitigation (287).

Conclusions and perspectives

Integrated summary: the special position of $\gamma\delta$ T cells in immunological strategy

$\gamma\delta$ T cells stand apart from the conventional $\alpha\beta$ lineage because they operate independently of MHC constraints and directly recognize stress or malignant signals (288–290). This independence grants them the quality of immediate response and the vigor of innate defense, while also bearing adaptive potential. Their distribution is not uniform: V δ 1+ $\gamma\delta$ T cells guard the mucosa and epithelial outposts of the body, whereas circulating V γ 9V δ 2+ $\gamma\delta$ T cells prove powerful against hematologic malignancies (118, 291, 292). In practice, clinical evidence has shown that in ALL and AML, infiltration by these cells correlates with improved survival (293–295).

Translational vision: broadening application from laboratory insight

Antigens such as CD123 become vulnerable to $\gamma\delta$ T cell targeting even when tumor HLA-I is lost (296–298). Clinical studies suggest that patients with post-transplant $\gamma\delta$ T-cell recovery >10% may have improved survival outcomes after transplant attain superior survival-proof of their graft-versus-leukemia role (299–301). In alliance with chemotherapy, radiotherapy, or checkpoint blockade, their effect is magnified (302–304). Further, bispecific antibodies and lymphodepletion schemes may rouse their force in refractory AML or CLL (305–307). Beyond malignancy, they can aid repair in aplastic anemia, though under other conditions they may intensify autoimmunity, as in hemolytic anemia (307–310). Thus, to distinguish subsets with single-cell sequencing is the path to stratified and precise application (311–313).

Strategic priorities and collaborative frameworks

To realize the clinical potential of $\gamma\delta$ T cells, standardized expansion protocols and precise subset stratification methods must be established, with their efficacy validated through rigorous clinical trials (314–316). Robust biomarkers must be developed to guide patient stratification, and multicenter collaborations are essential to accelerate clinical translation. Scholars and clinicians should collaborate to bridge the gap between basic immunobiology and clinical applications, effectively overcoming key barriers such as functional exhaustion, persistent *in vivo* limitations, and production variability.

Author contributions

XZ: Writing – original draft, Data curation, Investigation, Conceptualization, Writing – review & editing. MZ: Visualization, Writing – review & editing, Supervision. ZZ: Supervision, Writing – review & editing, Funding acquisition. GZ: Project administration, Supervision, Writing – review & editing, Conceptualization.

Funding

The author(s) declared financial support was received for this work and/or its publication. This work was supported by the Liaoning Provincial Applied Basic Research Program (Grant No. 2023JH2/101300037).

Acknowledgments

Figures in this work were created By [BioRender.com](https://www.biorender.com). We are grateful for the effort and support of Department of Hematology of The Fourth Affiliated Hospital of China Medical University.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. During the preparation of this work, the authors have not utilised generative artificial intelligence such as ChatGPT (OpenAI) for content creation or organisation. The authors have undertaken the necessary review and editing of the content and bear full responsibility for the contents of this publication.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

- Al-Samkari H. 2025 update on clinical trials in immune thrombocytopenia. *Am J Hematol.* (2024) 99:2178–90. doi: 10.1002/ajh.27448
- McReynolds LJ, Rafati M, Wang Y, Ballew BJ, Kim J, Williams VV, et al. Genetic testing in severe aplastic anemia is required for optimal hematopoietic cell transplant outcomes. *Blood.* (2022) 140:909–21. doi: 10.1182/blood.2022016508
- Shah S, Padrnos L. Complications of autoimmune hemolytic anemia. *Hematol Oncol Clin North Am.* (2022) 36:353–63. doi: 10.1016/j.hoc.2021.12.003
- Sallman DA, Stahl M. TP53-mutated acute myeloid leukemia: how can we improve outcomes? *Blood.* (2025) 145:2828–33. doi: 10.1182/blood.2024024245
- Green SD, Wang ES. How I treat secondary acute myeloid leukemia. *Blood.* (2025) 145:1260–72. doi: 10.1182/blood.2024024011
- Malard F, Mohty M. Acute lymphoblastic leukaemia. *Lancet.* (2020) 395:1146–62. doi: 10.1016/s0140-6736(19)33018-1
- Hampel PJ, Parikh SA. Chronic lymphocytic leukemia treatment algorithm 2022. *Blood Cancer J.* (2022) 12:161. doi: 10.1038/s41408-022-00756-9
- McCurdy A, Mian H, LeBlanc R, Jimenez-Zepeda VH, Su J, Masih-Khan E, et al. Redefining attrition in multiple myeloma (MM): a Canadian Myeloma Research Group (CMRG) analysis. *Blood Cancer J.* (2023) 13:111. doi: 10.1038/s41408-023-00883-x
- Sauerer T, Velazquez GF, Schmid C. Relapse of acute myeloid leukemia after allogeneic stem cell transplantation: immune escape mechanisms and current implications for therapy. *Mol Cancer.* (2023) 22:180. doi: 10.1186/s12943-023-01889-6
- The Lancet H. New insights into pregnancy after stem-cell transplantation. *Lancet Haematol.* (2024) 11:e629. doi: 10.1016/s2352-3026(24)00253-9
- Kong X, Xu Z, Wu Y, Tang X, Xue S, Miao M, et al. Increased Epstein-Barr virus reactivation following prophylaxis for cytomegalovirus infection after haploidentical haematopoietic stem cell transplantation. *J Hematol Oncol.* (2024) 17:94. doi: 10.1186/s13045-024-01612-y
- Mensurado S, Blanco-Dominguez R, Silva-Santos B. The emerging roles of gammadelta T cells in cancer immunotherapy. *Nat Rev Clin Oncol.* (2023) 20:178–91. doi: 10.1038/s41571-022-00722-1
- Yu X, Song L, Cen L, Cao B, Tao R, Shen Y, et al. Pan-cancer gammadelta TCR analysis uncovers clonotype diversity and prognostic potential. *Cell Rep Med.* (2024) 5:101764. doi: 10.1016/j.xcrm.2024.101764
- Kornberg A, Botella T, Moon CS, Rao S, Gelbs J, Cheng L, et al. Gluten induces rapid reprogramming of natural memory alphabeta and gammadelta intraepithelial T cells to induce cytotoxicity in celiac disease. *Sci Immunol.* (2023) 8:eadf4312. doi: 10.1126/sciimmunol.adf4312
- Wang L, Li J, Xuan Y, Zhang J, Wang X, Hu W, et al. Prospects for gammadelta T cells and chimeric antigen receptor gammadelta T cells in cancer immunotherapy. *Front Immunol.* (2025) 16:1554541. doi: 10.3389/fimmu.2025.1554541
- Patra S, Ghosal S, Shand H, Mondal R, Rath A, Kumar Jana S, et al. Function of gamma delta (gammadelta) T cell in cancer with special emphasis on cervical cancer. *Hum Immunol.* (2023) 84:110724. doi: 10.1016/j.humimm.2023.110724
- Hoeres T, Holzmann E, Smetak M, Birkmann J, Wilhelm M. PD-1 signaling modulates interferon-gamma production by Gamma Delta (gammadelta) T-Cells in response to leukemia. *Oncoimmunology.* (2019) 8:1550618. doi: 10.1080/2162402x.2018.1550618
- Zhang M, Li D, Zhu J, Xia X, Zhang H, Wu J, et al. IL-27 disturbs lipid metabolism and restrains mitochondrial activity to inhibit gammadelta T17 cell-mediated skin inflammation. *Cell Death Dis.* (2024) 15:491. doi: 10.1038/s41419-024-06887-0
- Rancan C, Arias-Badia M, Dogra P, Chen B, Aran D, Yang H, et al. Exhausted intratumoral Vdelta2(-) gammadelta T cells in human kidney cancer retain effector function. *Nat Immunol.* (2023) 24:612–24. doi: 10.1038/s41590-023-01448-7
- Cortes-Selva D, Dasgupta B, Singh S, Grewal IS. Innate and innate-like cells: the future of chimeric antigen receptor (CAR) cell therapy. *Trends Pharmacol Sci.* (2021) 42:45–59. doi: 10.1016/j.tips.2020.11.004
- Ferry GM, Agbuduwe C, Forrester M, Dunlop S, Chester K, Fisher J, et al. A simple and robust single-step method for CAR-Vdelta1 gammadeltaT cell expansion and transduction for cancer immunotherapy. *Front Immunol.* (2022) 13:863155. doi: 10.3389/fimmu.2022.863155
- Willcox CR, Mohammed F, Willcox BE. The distinct MHC-unrestricted immunobiology of innate-like and adaptive-like human gammadelta T cell subsets-Nature's CAR-T cells. *Immunol Rev.* (2020) 298:25–46. doi: 10.1111/imr.12928
- Moravec Z, Zhao Y, Voogd R, Cook DR, Kinrot S, Capra B, et al. Discovery of tumor-reactive T cell receptors by massively parallel library synthesis and screening. *Nat Biotechnol.* (2025) 43:214–22. doi: 10.1038/s41587-024-02210-6
- He J, Xiong X, Yang H, Li D, Liu X, Li S, et al. Defined tumor antigen-specific T cells potentiate personalized TCR-T cell therapy and prediction of immunotherapy response. *Cell Res.* (2022) 32:530–42. doi: 10.1038/s41422-022-00627-9
- Hsu J, Donahue RN, Katragadda M, Lowry J, Huang W, Srinivasan K, et al. A T cell receptor beta chain-directed antibody fusion molecule activates and expands subsets of T cells to promote antitumor activity. *Sci Transl Med.* (2023) 15:ead0258. doi: 10.1126/scitranslmed.adi0258
- Huang SW, Pan CM, Lin YC, Chen MC, Chen Y, Jan CI, et al. BiTE-secreting CAR-gammadeltaT as a dual targeting strategy for the treatment of solid tumors. *Adv Sci (Weinh).* (2023) 10:e2206856. doi: 10.1002/advs.202206856
- Makkouk A, Yang XC, Barca T, Lucas A, Turkoz M, Wong JTS, et al. Off-the-shelf Vdelta1 gamma delta T cells engineered with glypican-3 (GPC-3)-specific chimeric antigen receptor (CAR) and soluble IL-15 display robust antitumor efficacy against hepatocellular carcinoma. *J Immunother Cancer.* (2021) 9:e003441. doi: 10.1136/jitc-2021-003441
- Park JH, Lee HK. Function of gammadelta T cells in tumor immunology and their application to cancer therapy. *Exp Mol Med.* (2021) 53:318–27. doi: 10.1038/s12276-021-00576-0
- Mariuzza RA, Wu D, Pierce BG. Structural basis for T cell recognition of cancer neoantigens and implications for predicting neopeptide immunogenicity. *Front Immunol.* (2023) 14:1303304. doi: 10.3389/fimmu.2023.1303304
- Deseke M, Rampoldi F, Sandrock I, Borst E, Boning H, Ssebyatika GL, et al. A CMV-induced adaptive human Vdelta1+ gamma delta T cell clone recognizes HLA-DR. *J Exp Med.* (2022) 219(9):e20212525. doi: 10.1084/jem.20212525
- Bowen S, Sun P, Livak F, Sharrow S, Hodes RJ. A novel T cell subset with trans-rearranged Vgamma-Cbeta TCRs shows Vbeta expression is dispensable for lineage choice and MHC restriction. *J Immunol.* (2014) 192:169–77. doi: 10.4049/jimmunol.1302398
- Pishesha N, Harmand TJ, Ploegh HL. A guide to antigen processing and presentation. *Nat Rev Immunol.* (2022) 22:751–64. doi: 10.1038/s41577-022-00707-2
- Shaikh SR, Edidin M. Polyunsaturated fatty acids, membrane organization, T cells, and antigen presentation. *Am J Clin Nutr.* (2006) 84:1277–83. doi: 10.1093/ajcn/84.6.1277
- Komori HK, Meehan TF, Havran WL. Epithelial and mucosal gamma delta T cells. *Curr Opin Immunol.* (2006) 18:534–8. doi: 10.1016/j.coi.2006.06.001
- Wen X, Nian S, Wei G, Kang P, Yang Y, Li L, et al. Changes in the phenotype and function of mucosal-associated invariant T cells in neutrophilic asthma. *Int Immunopharmacol.* (2022) 106:108606. doi: 10.1016/j.intimp.2022.108606
- Qiu Y, Yang Y, Yang H. The unique surface molecules on intestinal intraepithelial lymphocytes: from tethering to recognizing. *Dig Dis Sci.* (2014) 59:520–9. doi: 10.1007/s10620-013-2933-1
- Zhu D, Ren X, Xie W, Chen J, Liang S, Jiang M, et al. Potential of gamma/delta T cells for solid tumor immunotherapy. *Front Immunol.* (2024) 15:1466266. doi: 10.3389/fimmu.2024.1466266
- Rao A, Agrawal A, Borthakur G, Battula VL, Maiti A. Gamma delta T cells in acute myeloid leukemia: biology and emerging therapeutic strategies. *J Immunother Cancer.* (2024) 12(2):e007981. doi: 10.1136/jitc-2023-007981
- Bhat J, Placek K, Faissner S. Contemplating dichotomous nature of gamma delta T cells for immunotherapy. *Front Immunol.* (2022) 13:894580. doi: 10.3389/fimmu.2022.894580
- Almeida AR, Correia DV, Fernandes-Platzgummer A, da Silva CL, da Silva MG, Anjos DR, et al. Delta One T cells for immunotherapy of chronic lymphocytic leukemia: clinical-grade expansion/differentiation and preclinical proof of concept. *Clin Cancer Res.* (2016) 22:5795–804. doi: 10.1158/1078-0432.ccr-16-0597
- Fowler D, Barisa M, Southern A, Nattress C, Hawkins E, Vassalou E, et al. Payload-delivering engineered gammadelta T cells display enhanced cytotoxicity, persistence, and efficacy in preclinical models of osteosarcoma. *Sci Transl Med.* (2024) 16:eadg9814. doi: 10.1126/scitranslmed.adg9814
- Arias-Badia M, Chang R, Fong L. gammadelta T cells as critical anti-tumor immune effectors. *Nat Cancer.* (2024) 5:1145–57. doi: 10.1038/s43018-024-00798-x
- Zheng Z, Wiedner T, Mauerer B, Schäfer L, Kesselring R, Braumüller H. T Cells in Colorectal Cancer: Unravelling the Function of Different T Cell Subsets in the Tumor Microenvironment. *Int J Mol Sci.* (2023) 24(14):11673. doi: 10.3390/ijms241411673
- Sullivan ZA, Khoury-Hanold W, Lim J, Smillie C, Biton M, Reis BS, et al. gamma delta T cells regulate the intestinal response to nutrient sensing. *Science.* (2021) 371(6535):eaba8310. doi: 10.1126/science.aba8310
- Xiao Z, Wang S, Luo L, Lv W, Feng P, Sun Y, et al. Lkb1 orchestrates gammadelta T-cell metabolic and functional fitness to control IL-17-mediated autoimmune hepatitis. *Cell Mol Immunol.* (2024) 21:546–60. doi: 10.1038/s41423-024-01163-9
- Torres-Hernandez A, Wang W, Nikiforov Y, Tejada K, Torres L, Kalabin A, et al. gammadelta T cells promote steatohepatitis by orchestrating innate and adaptive immune programming. *Hepatology.* (2020) 71:477–94. doi: 10.1002/hep.30952
- Sportoletti P, Sorcini D, Falini B. BCOR gene alterations in hematologic diseases. *Blood.* (2021) 138:2455–68. doi: 10.1182/blood.2021010958
- Zhao EJ, Cheng CV, Mattman A, Chen LYC. Polyclonal hypergammaglobulinaemia: assessment, clinical interpretation, and management. *Lancet Haematol.* (2021) 8:e365–e75. doi: 10.1016/s2352-3026(21)00056-9

49. Snowden JA, Sanchez-Ortega I, Corbacioglu S, Basak GW, Chabannon C, de la Camara R, et al. Indications for haematopoietic cell transplantation for haematological diseases, solid tumours and immune disorders: current practice in Europe, 2022. *Bone Marr Transplant.* (2022) 57:1217–39. doi: 10.1038/s41409-022-01691-w
50. Huang FT, Sun J, Zhang L, He X, Zhu YH, Dong HJ, et al. Role of SIRT1 in hematologic Malignancies. *J Zhejiang Univ-Sci B.* (2019) 20:391–8. doi: 10.1631/jzus.b1900148
51. Parrondo RD, Iqbal M, Von Roemeling R, Von Roemeling C, Tun HW. IRAK-4 inhibition: emavusertib for the treatment of lymphoid and myeloid Malignancies. *Front Immunol.* (2023) 14:1239082. doi: 10.3389/fimmu.2023.1239082
52. Larson RA. Etiology and management of therapy-related myeloid leukemia. *Hematology Am Soc Hematol Educ Program.* (2007) 2007:453–9. doi: 10.1182/asheducation-2007.1.453
53. Mangaonkar AA, Patnaik MM. Patterns of care and survival for elderly acute myeloid leukemia-challenges and opportunities. *Curr Hematol Malig Rep.* (2017) 12:290–9. doi: 10.1007/s11899-017-0388-8
54. Laubach J, Rao AV. Current and emerging strategies for the management of acute myeloid leukemia in the elderly. *Oncologist.* (2008) 13:1097–108. doi: 10.1634/theoncologist.2008-0100
55. Thol F, Ganser A. Treatment of relapsed acute myeloid leukemia. *Curr Treat Opt Oncol.* (2020) 21:66. doi: 10.1007/s11864-020-00765-5
56. Zarnegar-Lumley S, Caldwell KJ, Rubnitz JE. Relapsed acute myeloid leukemia in children and adolescents: current treatment options and future strategies. *Leukemia.* (2022) 36:1951–60. doi: 10.1038/s41375-022-01619-9
57. Piel FB, Rees DC, DeBaun MR, Nnodu O, Ranque B, Thompson AA, et al. Defining global strategies to improve outcomes in sickle cell disease: a Lancet Haematology Commission. *Lancet Haematol.* (2023) 10:e633–86. doi: 10.1016/s2352-3026(23)00096-0
58. Nelson MH, Paulos CM. Novel immunotherapies for hematologic Malignancies. *Immunol Rev.* (2015) 263:90–105. doi: 10.1111/imr.12245
59. Varga EA, Kerlin BA, Wurster MW. Social and ethical controversies in thrombophilia testing and update on genetic risk factors for venous thromboembolism. *Semin Thromb Hemost.* (2008) 34:549–61. doi: 10.1055/s-0028-1103366
60. Collaborators GBDSCD. Global, regional, and national prevalence and mortality burden of sickle cell disease, 2000–2021: a systematic analysis from the Global Burden of Disease Study 2021. *Lancet Haematol.* (2023) 10:e585–99. doi: 10.2139/ssrn.5381800
61. Osaki K, Morishita S, Takami S, Sakai Y, Kamimura A, Shindo A, et al. Quality of life of patients with hematological Malignancies and factors affecting health state utility values. *Supp Care Cancer.* (2022) 30:5319–27. doi: 10.1007/s00520-022-06958-y
62. Prabakaran A, Durrani J, Coelho-Da Silva J, Shalhoub R, Lotter J, Rios O, et al. Safety and efficacy of immunosuppressive therapy for elderly patients with severe aplastic anaemia. *Br J Haematol.* (2024) 205:1170–9. doi: 10.1111/bjh.19648
63. Ifitkhar R, DeFilipp Z, DeZern AE, Pulsipher MA, Bejanyan N, Burroughs LM, et al. Allogeneic hematopoietic cell transplantation for the treatment of severe aplastic anemia: evidence-based guidelines from the American Society for Transplantation and Cellular Therapy. *Transplant Cell Ther.* (2024) 30:1155–70. doi: 10.1016/j.jctct.2024.09.017
64. Tilly H, Morschhauser F, Sehn LH, Friedberg JW, Trnny M, Sharman JP, et al. Polatuzumab vedotin in previously untreated diffuse large B-cell lymphoma. *N Engl J Med.* (2022) 386:351–63. doi: 10.1056/nejmoa2115304
65. Al-Sarayfi D, Brink M, Chamuleau MED, Brouwer R, van Rijn RS, Issa D, et al. R-miniCHOP versus R-CHOP in elderly patients with diffuse large B-cell lymphoma: A propensity matched population-based study. *Am J Hematol.* (2024) 99(2):216–22. doi: 10.1002/ajh.27151
66. Wilson WH, Wright GW, Huang DW, Hodgkinson B, Balasubramanian S, Fan Y, et al. Effect of ibrutinib with R-CHOP chemotherapy in genetic subtypes of DLBCL. *Cancer Cell.* (2021) 39:1643–53.e3. doi: 10.1016/j.ccell.2021.10.006
67. Grosicki S, Simonova M, Spicka I, Pour L, Kriachok I, Gavriatopoulou M, et al. Once-per-week selinexor, bortezomib, and dexamethasone versus twice-per-week bortezomib and dexamethasone in patients with multiple myeloma (BOSTON): a randomised, open-label, phase 3 trial. *Lancet.* (2020) 396:1563–73. doi: 10.1016/s0140-6736(20)32292-3
68. Sogbein O, Paul P, Umar M, Chaari A, Batuman V, Upadhyay R. Bortezomib in cancer therapy: mechanisms, side effects, and future proteasome inhibitors. *Life Sci.* (2024) 358:123125. doi: 10.1016/j.lfs.2024.123125
69. Zhou X, Han S, Cebulla N, Haertle L, Steinhart MJ, Schirmer D, et al. Bortezomib induced peripheral neuropathy and single nucleotide polymorphisms in PKNOX1. *biomark Res.* (2023) 11:52. doi: 10.1186/s40364-023-00490-9
70. Keller CW, Sina C, Kotur MB, Ramelli G, Mundt S, Quast I, et al. ATG-dependent phagocytosis in dendritic cells drives myelin-specific CD4(+) T cell pathogenicity during CNS inflammation. *Proc Natl Acad Sci USA.* (2017) 114:E11228–37. doi: 10.1073/pnas.1713664114
71. van den Hoogen MW, Hoitsma AJ, Hilbrands LB. Anti-T-cell antibodies for the treatment of acute rejection after renal transplantation. *Expert Opin Biol Ther.* (2012) 12:1031–42. doi: 10.1517/14712598.2012.689278
72. Katano K, Nakanuma S, Araki T, Kato K, Sugita H, Gabata R, et al. Predictors of portal vein thrombosis after simultaneous hepatectomy and splenectomy: a single-center retrospective study. *Asian J Surg.* (2024) 47:5137–42. doi: 10.1016/j.asjsur.2024.07.131
73. Iolascon A, Andolfo I, Barcellini W, Corcione F, Garcon L, De Franceschi L, et al. Recommendations regarding splenectomy in hereditary hemolytic anemias. *Haematologica.* (2017) 102:1304–13. doi: 10.3324/haematol.2016.161166
74. Groarke EM, Patel BA, Gutierrez-Rodriguez F, Rios O, Lotter J, Baldoni D, et al. Eltrombopag added to immunosuppression for children with treatment-naïve severe aplastic anaemia. *Br J Haematol.* (2021) 192:605–14. doi: 10.1111/bjh.17232
75. Ejaz S, Ejaz S, Shahid R, Noor T, Shabbir S, Imran M. Chitosan-curcumin complexation to develop functionalized nanosystems with enhanced antimicrobial activity against hetero-resistant gastric pathogen. *Int J Biol Macromol.* (2022) 204:540–54. doi: 10.1016/j.jbiomac.2022.02.039
76. Inamoto Y, Lee SJ. Late effects of blood and marrow transplantation. *Haematologica.* (2017) 102:614–25. doi: 10.3324/haematol.2016.150250
77. Ballen K, Wang T, He N, Knight JM, Hong S, Frangoul H, et al. Impact of race and ethnicity on outcomes after umbilical cord blood transplantation. *Transplant Cell Ther.* (2024) 30:1027:e1–e14. doi: 10.1016/j.jctct.2024.07.009
78. Collaborators GBDVC. Global, regional, and national trends in routine childhood vaccination coverage from 1980 to 2023 with forecasts to 2030: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet.* (2025) 406:235–60. doi: 10.2139/ssrn.5265539
79. Xiao Y, Yu D. Tumor microenvironment as a therapeutic target in cancer. *Pharmacol Ther.* (2021) 221:107753. doi: 10.1016/j.pharmthera.2020.107753
80. Bader JE, Voss K, Rathmell JC. Targeting metabolism to improve the tumor microenvironment for cancer immunotherapy. *Mol Cell.* (2020) 78:1019–33. doi: 10.1016/j.molcel.2020.05.034
81. Gong Y, Jiang X, Zheng Y, Li Y, Bao X, Zhu W, et al. HLA alleles, blocks, and haplotypes associated with the hematological diseases of AML, ALL, MDS, and AA in the Han population of Southeastern China. *Chin Med J (Engl).* (2025) 138:877–9. doi: 10.1097/cm9.0000000000003521
82. Lee KH, Yoon SR, Gong JR, Choi EJ, Kim HS, Park CJ, et al. The infusion of ex vivo, interleukin-15 and -21-activated donor NK cells after haploidentical HCT in high-risk AML and MDS patients—a randomized trial. *Leukemia.* (2023) 37:807–19. doi: 10.1038/s41375-023-01849-5
83. Weidmann E, Hinz T, Klein S, Schui DK, Harder S, Kriener S, et al. Cytotoxic hepatosplenic gamma delta T-cell lymphoma following acute myeloid leukemia bearing two distinct gamma chains of the T-cell receptor. Biologic and clinical features. *Haematologica.* (2000) 85:1024–31. doi: 10.1016/j.clml.2015.11.019
84. Uckun FM, Watts J. CD123-directed bispecific antibodies for targeting MDS clones and immunosuppressive myeloid-derived suppressor cells (MDSC) in high-risk adult MDS patients. *Front Aging.* (2021) 2:757276. doi: 10.3389/fragi.2021.757276
85. Khairallah C, Dechanet-Merville J, Capone M. Gammadelta T cell-mediated immunity to cytomegalovirus infection. *Front Immunol.* (2017) 8:105. doi: 10.3389/fimmu.2017.00105
86. Vitale C, Boccellato E, Comba L, Foglietta M, Lobetti-Bodoni C, Ferrero S, et al. Impact of Immune Parameters and Immune Dysfunctions on the Prognosis of Patients with Chronic Lymphocytic Leukemia. *Cancers (Basel).* (2021) 13(15):3856. doi: 10.3390/cancers13153856
87. Riganti C, Castella B, Massaia M. ABCA1, apoA-I, and BTN3A1: a legitimate ménage à trois in dendritic cells. *Front Immunol.* (2018) 9:1246. doi: 10.3389/fimmu.2018.01246
88. Fichtner AS, Karunakaran MM, Gu S, Boughter CT, Borowska MT, Starick L, et al. Alpaca (Vicugna pacos), the first nonprimate species with a phosphoantigen-reactive Vgamma9Vdelta2 T cell subset. *Proc Natl Acad Sci USA.* (2020) 117:6697–707. doi: 10.1073/pnas.1909474117
89. Gruenbacher G, Nussbaumer O, Gander H, Steiner B, Leonhartsberger N, Thurnher M. Stress-related and homeostatic cytokines regulate Vgamma9Vdelta2 T-cell surveillance of mevalonate metabolism. *Oncoimmunology.* (2014) 3:e953410. doi: 10.4161/21624011.2014.953410
90. Willcox CR, Salim M, Begley CR, Karunakaran MM, Easton EJ, von Klopotek C, et al. Phosphoantigen sensing combines TCR-dependent recognition of the BTN3A IgV domain and germline interaction with BTN2A1. *Cell Rep.* (2023) 42:112321. doi: 10.1016/j.celrep.2023.112321
91. Zhu Y, Gao W, Zheng J, Bai Y, Tian X, Huang T, et al. Phosphoantigen-induced inside-out stabilization of butyrophilin receptor complexes drives dimerization-dependent gammadelta TCR activation. *Immunity.* (2025) 58:1646–59.e5. doi: 10.1016/j.immuni.2025.04.012
92. Cano CE, Pasero C, De Gassart A, Kerneur C, Gabriac M, Fullana M, et al. BTN2A1, an immune checkpoint targeting Vgamma9Vdelta2 T cell cytotoxicity against Malignant cells. *Cell Rep.* (2021) 36:109359. doi: 10.1016/j.celrep.2021.109359
93. Hsu UH, Chiang BL. Gammadelta T cells and allergic diseases. *Clin Rev Allergy Immunol.* (2023) 65:172–82. doi: 10.1007/s12016-023-08966-0

94. Cadoux M, Caruso S, Pham S, Gougelet A, Pophillat C, Riou R, et al. Expression of NKG2D ligands is downregulated by beta-catenin signalling and associates with HCC aggressiveness. *J Hepatol.* (2021) 74:1386–97. doi: 10.1016/j.jhep.2021.01.017
95. Suárez-Alvarez B, López-Vázquez A, Baltar JM, Ortega F, López-Larrea C. Potential role of NKG2D and its ligands in organ transplantation: new target for immunointervention. *Am J Transplant.* (2009) 9(2):251–7. doi: 10.1111/j.1600-6143.2008.02526.x
96. Lin F, Dai C, Ge X, Tang W, Lin Y, Wang Y, et al. Prognostic significance and functional implication of immune activating receptor NKG2D in gastric cancer. *Biochem Biophys Res Commun.* (2017) 487:619–24. doi: 10.1016/j.bbrc.2017.04.104
97. Tanaka Y. Cancer immunotherapy harnessing gammadelta T cells and programmed death-1. *Immunol Rev.* (2020) 298:237–53. doi: 10.1111/imr.12917
98. Hodgins NO, Wang JT, Al-Jamal KT. Nano-technology based carriers for nitrogen-containing bisphosphonates delivery as sensitizers of gammadelta T cells for anticancer immunotherapy. *Adv Drug Delivery Rev.* (2017) 114:143–60. doi: 10.1016/j.addr.2017.07.003
99. Clezardin P, Massia M. Nitrogen-containing bisphosphonates and cancer immunotherapy. *Curr Pharm Des.* (2010) 16:3007–2014. doi: 10.2174/138161210793563545
100. Hsu H, Boudova S, Mvula G, Divala TH, Rach D, Mungwira RG, et al. Age-related changes in PD-1 expression coincide with increased cytotoxic potential in Vdelta2 T cells during infancy. *Cell Immunol.* (2021) 359:104244. doi: 10.1016/j.cellimm.2020.104244
101. Zhou J, Zhang J, Tao L, Peng K, Zhang Q, Yan K, et al. Up-regulation of BTN3A1 on CD14(+) cells promotes Vgamma9Vdelta2 T cell activation in psoriasis. *Proc Natl Acad Sci USA.* (2022) 119:e2117523119. doi: 10.1073/pnas.2117523119
102. Domae E, Hirai Y, Ikeo T, Goda S, Shimizu Y. Cytokine-mediated activation of human ex vivo-expanded Vgamma9Vdelta2 T cells. *Oncotarget.* (2017) 8:45928–42. doi: 10.18632/oncotarget.17498
103. Ness-Schwickerath KJ, Morita CT. Regulation and function of IL-17A- and IL-22-producing gammadelta T cells. *Cell Mol Life Sci.* (2011) 68:2371–90. doi: 10.1007/s00181-011-0700-z
104. MaChado NR, Fagundes BO, Fernandes LA, de Oliveira ACP, Nukui Y, Casseb J, et al. Differential modulation of IL-4, IL-10, IL-17, and IFN-gamma production mediated by IgG from Human T-lymphotropic virus-1 (HTLV-1) infected patients on healthy peripheral T (CD4+, CD8+, and gammadelta) and B cells. *Front Med (Laus).* (2023) 10:1239706. doi: 10.3389/fmed.2023.1239706
105. Fagundes BO, de-Sousa TR, Victor JR. Gamma-delta ($\gamma\delta$) T cell-derived cytokines (IL-4, IL-17, IFN- γ and IL-10) and their possible implications for atopic dermatitis development. *Int J Dermatol.* (2023) 62(4):443–8. doi: 10.1111/ijd.16364
106. Paget C, Chow MT, Gherardin NA, Beavis PA, Uldrich AP, Duret H, et al. CD3bright signals on $\gamma\delta$ T cells identify IL-17A-producing V γ 6V δ 1+ T cells. *Immunol Cell Biol.* (2015) 93(2):198–212. doi: 10.1038/icb.2014.94
107. Wu H, Knight JF, Alexander SI. Regulatory gamma delta T cells in Heymann nephritis express an invariant Vgamma6/Vdelta1 with a canonical CDR3 sequence. *Eur J Immunol.* (2004) 34:2322–30. doi: 10.1002/eji.200324780
108. Allegra A, Casciaro M, Lo Presti E, Musolino C, Gangemi S. Harnessing Unconventional T Cells and Innate Lymphoid Cells to Prevent and Treat Hematological Malignancies: Prospects for New Immunotherapy. *Biomolecules.* (2022) 12(6):754. doi: 10.3390/biom12060754
109. Munoz-Ruiz M, Llorian M, D'Antuono R, Pavlova A, Mavrigiannaki AM, McKenzie D, et al. IFN-gamma-dependent interactions between tissue-intrinsic gammadelta T cells and tissue-infiltrating CD8 T cells limit allergic contact dermatitis. *J Allergy Clin Immunol.* (2023) 152:1520–40. doi: 10.1016/j.jaci.2023.07.015
110. Sumaria N, Fiala GJ, Inacio D, Curado-Avelar M, Cachucho A, Pinheiro R, et al. Perinatal thymic-derived CD8alpha-beta-expressing gammadelta T cells are innate IFN-gamma producers that expand in IL-7R-STAT5B-driven neoplasms. *Nat Immunol.* (2024) 25:1207–17. doi: 10.1038/s41590-024-01855-4
111. Duan Y, Li G, Xu M, Qi X, Deng M, Lin X, et al. CFTR is a negative regulator of gammadelta T cell IFN-gamma production and antitumor immunity. *Cell Mol Immunol.* (2021) 18:1934–44. doi: 10.1038/s41423-020-0499-3
112. Piepke M, Clausen BH, Ludewig P, Vienhues JH, Bedke T, Javidi E, et al. Interleukin-10 improves stroke outcome by controlling the detrimental Interleukin-17A response. *J Neuroinflamm.* (2021) 18:265. doi: 10.1186/s12974-021-02316-7
113. Cheng WC, Saleh F, Abuaisa Karim B, Hughes FJ, Taams LS. Comparative analysis of immune cell subsets in peripheral blood from patients with periodontal disease and healthy controls. *Clin Exp Immunol.* (2018) 194:380–90. doi: 10.1111/cei.13205
114. Hazeldine J, Lord JM. The impact of ageing on natural killer cell function and potential consequences for health in older adults. *Ageing Res Rev.* (2013) 12:1069–78. doi: 10.1016/j.arr.2013.04.003
115. Kawasaki N, Yamashita-Kashima Y, Fujimura T, Yoshiura S, Harada N, Kondoh O, et al. Resistance to obinituzumab-induced antibody-dependent cellular cytotoxicity caused by abnormal Fas signaling is overcome by combination therapies. *Mol Biol Rep.* (2022) 49:4421–33. doi: 10.1007/s11033-022-07280-w
116. Harly C, Guillaume Y, Nedellec S, Peigne CM, Monkkonen H, Monkkonen J, et al. Key implication of CD277/butyrophilin-3 (BTN3A) in cellular stress sensing by a major human gammadelta T-cell subset. *Blood.* (2012) 120:2269–79. doi: 10.1182/blood-2012-05-430470
117. Mamedov MR, Vedova S, Freimer JW, Sahu AD, Ramesh A, Arce MM, et al. CRISPR screens decode cancer cell pathways that trigger gammadelta T cell detection. *Nature.* (2023) 621:188–95. doi: 10.1038/s41586-023-06482-x
118. Halary F, Pitard V, Dlubek D, Krzysiek R, de la Salle H, Merville P, et al. Shared reactivity of Vdelta2(neg) gammadelta T cells against cytomegalovirus-infected cells and tumor intestinal epithelial cells. *J Exp Med.* (2005) 201:1567–78. doi: 10.1084/jem.20041851
119. Wu J, Groh V, Spies T. T cell antigen receptor engagement and specificity in the recognition of stress-inducible MHC class I-related chains by human epithelial gamma delta T cells. *J Immunol.* (2002) 169:1236–42. doi: 10.4049/jimmunol.169.3.1236
120. Qu X, Wang Z, Zhou T, Shan L. Determination of the molecular mechanism by which macrophages and gammadelta-T cells contribute to ZOL-induced ONJ. *Aging (Alb NY).* (2020) 12:20743–52. doi: 10.18632/aging.104006
121. Hudspeth K, Fogli M, Correia DV, Mikulak J, Roberto A, Della Bella S, et al. Engagement of NKp30 on Vdelta1 T cells induces the production of CCL3, CCL4, and CCL5 and suppresses HIV-1 replication. *Blood.* (2012) 119:4013–6. doi: 10.1182/blood-2011-11-390153
122. Vatter S, Schmid M, Gebhard C, Mirbeth C, Klobuch S, Rehli M, et al. In-vitro blockade of the CD4 receptor co-signal in antigen-specific T-cell stimulation cultures induces the outgrowth of potent CD4 independent T-cell effectors. *J Immunol Methods.* (2018) 454:80–5. doi: 10.1016/j.jim.2017.11.006
123. Duan M, Goswami S, Shi JY, Wu LJ, Wang XY, Ma JQ, et al. Activated and exhausted MAIT cells foster disease progression and indicate poor outcome in hepatocellular carcinoma. *Clin Cancer Res.* (2019) 25:3304–16. doi: 10.1158/1078-0432.ccr-18-3040
124. Renauer P, Park JJ, Bai M, Acosta A, Lee WH, Lin GH, et al. Immunogenetic metabolomics reveals key enzymes that modulate CAR T-cell metabolism and function. *Cancer Immunol Res.* (2023) 11:1068–74. doi: 10.1158/2326-6066.cir-22-0565
125. Kim KJ, Moon D, Kong SJ, Lee YS, Yoo Y, Kim S, et al. Antitumor effects of IL-12 and GM-CSF co-expressed in an engineered oncolytic HSV-1. *Gene Ther.* (2021) 28:186–98. doi: 10.1038/s41434-020-00205-x
126. King IL, Kroenke MA, Segal BM. GM-CSF-dependent, CD103+ dermal dendritic cells play a critical role in Th effector cell differentiation after subcutaneous immunization. *J Exp Med.* (2010) 207:953–61. doi: 10.1084/jem.20091844
127. Liontos LM, Dissanayake D, Ohashi PS, Weiss A, Dragone LL, McGlade CJ. The Src-like adaptor protein regulates GM-CSFR signaling and monocytic dendritic cell maturation. *J Immunol.* (2011) 186:1923–33. doi: 10.4049/jimmunol.0903292
128. Shang QN, Yu XX, Xu ZL, Chen YH, Han TT, Zhang YY, et al. Expanded clinical-grade NK cells exhibit stronger effects than primary NK cells against HCMV infection. *Cell Mol Immunol.* (2023) 20:895–907. doi: 10.1038/s41423-023-01046-5
129. Eriksson E, Milenova I, Wenthe J, Stahle M, Leja-Jarblad J, Ullenhag G, et al. Shaping the tumor stroma and sparking immune activation by CD40 and 4-1BB signaling induced by an armed oncolytic virus. *Clin Cancer Res.* (2017) 23:5846–57. doi: 10.1158/1078-0432.ccr-17-0285
130. Martin AJ, Clark M, Gojanovich G, Manzoof F, Miller K, Kline DE, et al. Anti-coreceptor therapy drives selective T cell egress by suppressing inflammation-dependent chemotactic cues. *JCI Insight.* (2016) 1:e87636. doi: 10.1172/jci.insight.87636
131. Ngo VN, Tang HL, Cyster JG. Epstein-Barr virus-induced molecule 1 ligand chemokine is expressed by dendritic cells in lymphoid tissues and strongly attracts naive T cells and activated B cells. *J Exp Med.* (1998) 188:181–91. doi: 10.1084/jem.188.1.181
132. Neves EGA, Koh CC, Lucinda PPD, Souza-Silva TG, Medeiros NI, Pantaleao A, et al. Blocking activation of CD4(-)CD8(-) T cells modulates their cytotoxic potential and decreases the expression of inflammatory and chemotactic receptors. *Clin Immunol.* (2023) 251:109331. doi: 10.1016/j.clim.2023.109331
133. Royset ES, Sahlin Pettersen HP, Xu W, Larbi A, Sandvik AK, Steigen SE, et al. Deep learning-based image analysis reveals significant differences in the number and distribution of mucosal CD3 and $\gamma\delta$ T cells between Crohn's disease and ulcerative colitis. *J Pathol Clin Res.* (2023) 9(1):18–31. doi: 10.1002/cjp.2301
134. Arnold S, Holl Ulrich K, Lamprecht P. Pathogenese der Großgefäßvaskulitiden. *Z Rheumatol.* (2020) 79(6):505–15. doi: 10.1007/s00393-020-00809-z
135. Dong C. Cytokine regulation and function in T cells. *Annu Rev Immunol.* (2021) 39:51–76. doi: 10.1146/annurev-immunol-061020-053702
136. Ribot JC, Lopes N, Silva-Santos B. Gammadelta T cells in tissue physiology and surveillance. *Nat Rev Immunol.* (2021) 21:221–32. doi: 10.1038/s41577-020-00452-4
137. Mills KHG. IL-17 and IL-17-producing cells in protection versus pathology. *Nat Rev Immunol.* (2023) 23:38–54. doi: 10.1038/s41577-022-00746-9
138. Cruz MS, Diamond A, Russell A, Jameson JM. Human alphabeta and gammadelta T cells in skin immunity and disease. *Front Immunol.* (2018) 9:1304. doi: 10.3389/fimmu.2018.01304

139. Li Z, Lu X, Hu Z, Luo Z, Jiang W, Wu H, et al. Syphilis infection differentially regulates the phenotype and function of gammadelta T cells in HIV-1-infected patients depends on the HIV-1 disease stage. *Front Immunol.* (2017) 8:991. doi: 10.3389/fimmu.2017.00991
140. Fakharian F, Thirugnanam S, Welsh DA, Kim WK, Rappaport J, Bittiger K, et al. The Role of Gut Dysbiosis in the Loss of Intestinal Immune Cell Functions and Viral Pathogenesis. *Microorganisms.* (2023) 11(7):1849. doi: 10.3390/microorganisms11071849
141. Reddy NR, Maachi H, Xiao Y, Simic MS, Yu W, Tonai Y, et al. Engineering synthetic suppressor T cells that execute locally targeted immunoprotective programs. *Science.* (2024) 386:eadi4793. doi: 10.1126/science.adi4793
142. Gu Q, Draheim M, Planchais C, He Z, Mu F, Gong S, et al. Intestinal newborn regulatory B cell antibodies modulate microbiota communities. *Cell Host Microbe.* (2024) 32:1787–804.e9. doi: 10.1016/j.chom.2024.08.010
143. Yi Y, He HW, Wang JX, Cai XY, Li YW, Zhou J, et al. The functional impairment of HCC-infiltrating gammadelta T cells, partially mediated by regulatory T cells in a TGFbeta- and IL-10-dependent manner. *J Hepatol.* (2013) 58:977–83. doi: 10.1016/j.jhep.2012.12.015
144. Zhang X, Zhu L, Zhang H, Chen S, Xiao Y. CAR-T cell therapy in hematological Malignancies: Current opportunities and challenges. *Front Immunol.* (2022) 13:927153. doi: 10.3389/fimmu.2022.927153
145. Tang L, Huang Z, Mei H, Hu Y. Immunotherapy in hematologic Malignancies: Achievements, challenges and future prospects. *Signal Transd Targ Ther.* (2023) 8:306. doi: 10.1038/s41392-023-01521-5
146. Yu S, Ren X, Li L. Myeloid-derived suppressor cells in hematologic Malignancies: Two sides of the same coin. *Exp Hematol Oncol.* (2022) 11:43. doi: 10.1186/s40164-022-00296-9
147. Laetsch TW, Maude SL, Rives S, Hiramatsu H, Bittencourt H, Bader P, et al. Three-year update of tisagenlecleucel in pediatric and young adult patients with relapsed/refractory acute lymphoblastic leukemia in the ELIANA trial. *J Clin Oncol.* (2023) 41:1664–9. doi: 10.1200/jco.22.00642
148. Brown PA, Ji L, Xu X, Devidas M, Hogan LE, Borowitz MJ, et al. Effect of postreinduction therapy consolidation with blinatumomab vs chemotherapy on disease-free survival in children, adolescents, and young adults with first relapse of B-cell acute lymphoblastic leukemia: A randomized clinical trial. *JAMA.* (2021) 325:833–42. doi: 10.1001/jama.2021.0669
149. Shah BD, Ghobadi A, Oluwole OO, Logan AC, Boissel N, Cassaday RD, et al. KTE-X19 for relapsed or refractory adult B-cell acute lymphoblastic leukaemia: Phase 2 results of the single-arm, open-label, multicentre ZUMA-3 study. *Lancet.* (2021) 398:491–502. doi: 10.1016/s0140-6736(21)01222-8
150. Nguyen PC, Manos K, Fong CY, Schwarzer AP, Tiong IS, Wei AH, et al. Outcomes of non-myeloablative allogeneic stem cell transplant in older patients with acute myeloid leukaemia in first remission. *Intern Med J.* (2021) 51:1954–8. doi: 10.1111/imj.15564
151. Keegan THM, Bleyer A, Rosenberg AS, Li Q, Goldfarb M. Second primary Malignant neoplasms and survival in adolescent and young adult cancer survivors. *JAMA Oncol.* (2017) 3:1554–7. doi: 10.1001/jamaoncol.2017.0465
152. Kreuzberger N, Damen JA, Trivella M, Estcourt LJ, Aldin A, Umlauff L, et al. Prognostic models for newly-diagnosed chronic lymphocytic leukaemia in adults: A systematic review and meta-analysis. *Cochrane Database Syst Rev.* (2020) 7:CD012022. doi: 10.1002/14651858.cd012022.pub2
153. Ahn IE, Brander DM, Ren Y, Zhou Y, Tyekucheva S, Walker HA, et al. Five-year follow-up of a phase 2 study of ibrutinib plus fludarabine, cyclophosphamide, and rituximab as initial therapy in CLL. *Blood Adv.* (2024) 8:832–41. doi: 10.1182/bloodadvances.2023011574
154. Hallek M. Chronic lymphocytic leukemia: 2025 update on the epidemiology, pathogenesis, diagnosis, and therapy. *Am J Hematol.* (2025) 100:450–80. doi: 10.1002/ajh.27546
155. Avet-Loiseau H, Davies FE, Samur MK, Corre J, D'Agostino M, Kaiser MF, et al. International Myeloma Society/International Myeloma Working Group consensus recommendations on the definition of high-risk multiple myeloma. *J Clin Oncol.* (2025) 43:2739–51. doi: 10.1200/jco-24-01893
156. Tang L, Zhang H, Zhou F, Wei Q, Du M, Wu J, et al. Targeting autophagy overcomes cancer-intrinsic resistance to CAR-T immunotherapy in B-cell Malignancies. *Cancer Commun (Lond).* (2024) 44:408–32. doi: 10.1002/cac2.12525
157. Abramson JS, Lunning M, Palomba ML. Chimeric antigen receptor T-cell therapies for aggressive B-cell lymphomas: Current and future state of the art. *Am Soc Clin Oncol Educ Book.* (2019) 39:446–53. doi: 10.1200/edbk_238693
158. Shah NN, Fry TJ. Mechanisms of resistance to CAR T cell therapy. *Nat Rev Clin Oncol.* (2019) 16:372–85. doi: 10.1038/s41571-019-0184-6
159. Xu X, Zu C, Zhang M, Xiao P, Hong R, Feng J, et al. HLA fully-mismatched sibling-derived CD7 CAR-T therapy bridging to haploidentical hematopoietic stem cell transplantation for hepatosplenic gammadelta T-cell lymphoma. *Cell Transplant.* (2023) 32:9636897231194265. doi: 10.1177/09636897231194265
160. Ruella M, Locke FL. Beat pediatric ALL MRD: CD28 CAR T and transplant. *Blood.* (2019) 134:2333–5. doi: 10.1182/blood.2019003821
161. Klyuchnikov E, Badbaran A, Massoud R, Fritsche-Friedland U, Janson D, Ayuk F, et al. Enhanced immune reconstitution of gammadelta T cells after allogeneic stem cell transplantation overcomes the negative impact of pretransplantation minimal residual disease-positive status in patients with acute myelogenous leukemia. *Transplant Cell Ther.* (2021) 27:841–50. doi: 10.1016/j.jct.2021.06.003
162. Madhok A, Bhat SA, Philip CS, Sureshbabu SK, Chiplunkar S, Galande S. Transcriptome signature of Vgamma9Vdelta2 T cells treated with phosphoantigens and Notch inhibitor reveals interplay between TCR and Notch signaling pathways. *Front Immunol.* (2021) 12:660361. doi: 10.3389/fimmu.2021.660361
163. Pistoia V, Tumino N, Vacca P, Veneziani I, Moretta A, Locatelli F, et al. Human gammadelta T-cells: From surface receptors to the therapy of high-risk leukemias. *Front Immunol.* (2018) 9:984. doi: 10.3389/fimmu.2018.00984
164. Lu H, Shi T, Wang M, Li X, Gu Y, Zhang X, et al. B7-H3 inhibits the IFN-gamma-dependent cytotoxicity of Vgamma9Vdelta2 T cells against colon cancer cells. *Oncoimmunology.* (2020) 9:1748991. doi: 10.1080/2162402x.2020.1748991
165. Minculescu L, Sengelov H, Marquart HV, Ryder LP, Fischer-Nielsen A, Hastrup E. Granulocyte colony-stimulating factor effectively mobilizes TCR gammadelta and NK cells providing an allograft potentially enhanced for the graft-versus-leukemia effect for allogeneic stem cell transplantation. *Front Immunol.* (2021) 12:625165. doi: 10.3389/fimmu.2021.625165
166. Scheper W, van Dorp S, Kersting S, Pietersma F, Lindemans C, Hol S, et al. gammadeltaT cells elicited by CMV reactivation after allo-SCT cross-recognize CMV and leukemia. *Leukemia.* (2013) 27:1328–38. doi: 10.1038/leu.2012.374
167. Bertaina A, Zoroli A, Petretto A, Barbarito G, Inglese E, Merli P, et al. Zoledronic acid boosts gammadelta T-cell activity in children receiving alphabeta(+) T and CD19 (+) cell-depleted grafts from an HLA-haplo-identical donor. *Oncoimmunology.* (2017) 6:e1216291. doi: 10.1182/blood.v128.22.5771.5771
168. Perna F, Berman SH, Soni RK, Mansilla-Soto J, Eyquem J, Hamieh M, et al. Integrating proteomics and transcriptomics for systematic combinatorial chimeric antigen receptor therapy of AML. *Cancer Cell.* (2017) 32:506–19.e5. doi: 10.1016/j.ccell.2017.09.004
169. Kong Y, Yoshida S, Saito Y, Doi T, Nagatoshi Y, Fukata M, et al. CD34+CD38 +CD19+ as well as CD34+CD38-CD19+ cells are leukemia-initiating cells with self-renewal capacity in human B-precursor ALL. *Leukemia.* (2008) 22:1207–13. doi: 10.1038/leu.2008.83
170. Wittebol S, Raymakers R, van de Locht L, Mensink E, de Witte T. In AML t(8;21) colony growth of both leukemic and residual normal progenitors is restricted to the CD34+, lineage-negative fraction. *Leukemia.* (1998) 12:1782–8. doi: 10.1038/sj.leu.2401178
171. Garaud S, Morva A, Lemoine S, Hillion S, Bordron A, Pers JO, et al. CD5 promotes IL-10 production in chronic lymphocytic leukemia B cells through STAT3 and NFAT2 activation. *J Immunol.* (2011) 186:4835–44. doi: 10.4049/jimmunol.1003050
172. Barcellini W, Montesano R, Clerici G, Zaninoni A, Imperiali FG, Calori R, et al. In vitro production of anti-RBC antibodies and cytokines in chronic lymphocytic leukemia. *Am J Hematol.* (2002) 71(3):177–83. doi: 10.1002/ajh.10210
173. Poggi A, Venturino C, Catellani S, Clavio M, Miglino M, Gobbi M, et al. Vdelta1 T lymphocytes from B-CLL patients recognize ULBP3 expressed on leukemic B cells and up-regulated by trans-retinoic acid. *Cancer Res.* (2004) 64:9172–9. doi: 10.1158/0008-5472.can.04-2417
174. King LA, Toffoli EC, Veth M, Iglesias-Guimaraes V, Slot MC, Amsen D, et al. A bispecific gammadelta T-cell engager targeting EGFR activates a potent Vgamma9Vdelta2 T cell-mediated immune response against EGFR-expressing tumors. *Cancer Immunol Res.* (2023) 11:1237–52. doi: 10.1158/2326-6066.cir-23-0189
175. Li X, Lu H, Gu Y, Zhang X, Zhang G, Shi T, et al. Tim-3 suppresses the killing effect of Vgamma9Vdelta2 T cells on colon cancer cells by reducing perforin and granzyme B expression. *Exp Cell Res.* (2020) 386:111719. doi: 10.1016/j.yexcr.2019.111719
176. Castella B, Riganti C, Massaia M. Metabolic approaches to rescue antitumor Vgamma9Vdelta2 T-cell functions in myeloma. *Front Biosci (Landmark Ed).* (2020) 25(1):69–105. doi: 10.2741/4795
177. Vetter VK, Benoit T, Rahmani-Khajouei S, Pantic M, Wickenhauser C, Federmann B, et al. Nodal gamma-delta T-cell lymphoma: expression of TFH-markers causing diagnostic difficulties in a newly described entity. *Virchows Arch.* (2025) 487(5):1467–73. doi: 10.1007/s00428-025-04268-2
178. King LA, Veth M, Iglesias-Guimaraes V, Blijdorp I, Kloosterman J, Vis AN, et al. Leveraging Vgamma9Vdelta2 T cells against prostate cancer through a VHH-based PSMA-Vdelta2 bispecific T cell engager. *iScience.* (2024) 27:111289. doi: 10.1016/j.isci.2024.111289
179. Salaroglio IC, Belisario DC, Akman M, La Vecchia S, Godel M, Anobile DP, et al. Mitochondrial ROS drive resistance to chemotherapy and immune-killing in hypoxic non-small cell lung cancer. *J Exp Clin Cancer Res.* (2022) 41:243. doi: 10.1186/s13046-022-02447-6
180. Shah NN, Lee DW, Yates B, Yuan CM, Shalabi H, Martin S, et al. Long-term follow-up of CD19-CAR T-cell therapy in children and young adults with B-ALL. *J Clin Oncol.* (2021) 39:1650–9. doi: 10.1182/hematology.2023000422

181. Briski R, Garcia-Manero G, Kantarjian H, Ravandi F. The history of oral decitabine/cedazuridine and its potential role in acute myeloid leukemia. *Ther Adv Hematol*. (2023) 14:20406207231205429. doi: 10.1177/20406207231205429
182. Tausch E, Schneider C, Stilgenbauer S. Risk-stratification in frontline CLL therapy: Standard of care. *Hematol Am Soc Hematol Educ Program*. (2024) 2024:457–66. doi: 10.1182/hematology.2024000656
183. Ma H, Padmanabhan IS, Parmar S, Gong Y. Targeting CLL-1 for acute myeloid leukemia therapy. *J Hematol Oncol*. (2019) 12:41. doi: 10.1186/s13045-019-0726-5
184. Jaeger-Ruckstuhl CA, Specht JM, Voutsinas JM, MacMillan HR, Wu QV, Muhunthan V, et al. Phase I study of ROR1-specific CAR-T cells in advanced hematopoietic and epithelial Malignancies. *Clin Cancer Res*. (2025) 31:503–14. doi: 10.1158/1078-0432.ccr-24-2172
185. Zhai X, You F, Xiang S, Jiang L, Chen D, Li Y, et al. MUC1-Tn-targeting chimeric antigen receptor-modified V γ 9V δ 2 T cells with enhanced antigen-specific anti-tumor activity. *Am J Cancer Res*. (2021) 11(1):79–91. doi: 10.21147/ajcr.0110806
186. Yu B, Jiang T, Liu D. BCMA-targeted immunotherapy for multiple myeloma. *J Hematol Oncol*. (2020) 13:125. doi: 10.1186/s13045-020-00962-7
187. Lee H, Durante M, Skerget S, Vishwamitra D, Benaoudia S, Ahn S, et al. Impact of soluble BCMA and non-T-cell factors on refractoriness to BCMA-targeting T-cell engagers in multiple myeloma. *Blood*. (2024) 144:2637–51. doi: 10.1182/blood.2024026212
188. Mato AR, Shah NN, Jurczak W, Cheah CY, Pagel JM, Woyach JA, et al. Pirtobrutinib in relapsed or refractory B-cell Malignancies (BRUIN): A phase 1/2 study. *Lancet*. (2021) 397:892–901. doi: 10.1016/s0140-6736(21)00224-5
189. Mei MG, Lee HJ, Palmer JM, Chen R, Tsai NC, Chen L, et al. Response-adapted anti-PD-1-based salvage therapy for Hodgkin lymphoma with nivolumab alone or in combination with ICE. *Blood*. (2022) 139:3605–16. doi: 10.1182/ashsap9.chapter43
190. Nishimoto KP, Barca T, Azameera A, Makkouk A, Romero JM, Bai L, et al. Allogeneic CD20-targeted gammadelta T cells exhibit innate and adaptive antitumor activities in preclinical B-cell lymphoma models. *Clin Transl Immunol*. (2022) 11:e1373. doi: 10.1002/cti2.1373
191. Bhagwat AS, Torres L, Shestova O, Shestov M, Mellors PW, Fisher HR, et al. Cytokine-mediated CAR T therapy resistance in AML. *Nat Med*. (2024) 30:3697–708. doi: 10.1038/s41591-024-03271-5
192. Zeng Z, Roobrouck A, Deschamps G, Bonnevaux H, Guerif S, De Brabandere V, et al. Dual-targeting CD33/CD123 NANOBODY T-cell engager with potent anti-AML activity and good safety profile. *Blood Adv*. (2024) 8:2059–73. doi: 10.1182/bloodadvances.2023011858
193. Stocker N, Gaugler B, Labopin M, Farge A, Ye Y, Ricard L, et al. High-dose post-transplant cyclophosphamide impairs gammadelta T-cell reconstitution after haploidentical haematopoietic stem cell transplantation using low-dose antithymocyte globulin and peripheral blood stem cell graft. *Clin Transl Immunol*. (2020) 9:e1171. doi: 10.1002/cti2.1171
194. Nattress CB, O'Sullivan R, Fowler D, Hutton C, Vlckova P, Sufi J, et al. Phenocapting Reveals Multimodal $\gamma\delta$ T-cell Cytotoxicity as a Strategy to Overcome Cancer Cell-Mediated Immunomodulation. *Cancer Res*. (2025) 85(22):4415–32. doi: 10.1158/0008-5472.CAN-25-1890
195. Zhang Y, Liu J, Yao J, Yuan N, Xu R. Mechanisms and applications of $\gamma\delta$ T cells in anti-tumor immunity. *Cancer Ctrl*. (2025) 32:10732748251399242. doi: 10.1177/10732748251399242
196. Durand A, Porte S, Xing E, Wu C, Le Bon A, Auffray C, et al. Aging Reshapes $\gamma\delta$ T-Cell Immunity Through a Type I Interferon-Foxo1 Axis. *Aging Cell*. (2026) 25(2):e70389. doi: 10.1111/accel.70389
197. Wu K, Feng J, Xiu Y, Li Z, Lin Z, Zhao H, et al. Vdelta2 T cell subsets, defined by PD-1 and TIM-3 expression, present varied cytokine responses in acute myeloid leukemia patients. *Int Immunopharmacol*. (2020) 80:106122. doi: 10.1016/j.intimp.2019.106122
198. Cimini E, Tartaglia E, Messina F, Coppola A, Mazzotta V, Tempestilli M, et al. Vdelta2 T-cells response in people with Mpox infection: A three-month longitudinal assessment. *Emerg Microbes Infect*. (2025) 14:2455585. doi: 10.1080/22221751.2025.2455585
199. Meggyes M, Lajko A, Palkovics T, Totsimon A, Illes Z, Szereday L, et al. Feto-maternal immune regulation by TIM-3/galectin-9 pathway and PD-1 molecule in mice at day 14.5 of pregnancy. *Placenta*. (2015) 36:1153–60. doi: 10.1016/j.placenta.2015.07.124
200. Cha L, Jones AP, Trend S, Byrne SN, Fabis-Pedrini MJ, Carroll WM, et al. Tryptophan and arginine catabolic enzymes and regulatory cytokines in clinically isolated syndrome and multiple sclerosis. *Clin Transl Immunol*. (2018) 7:e1037. doi: 10.1002/cti2.1037
201. Andersen MH. The T-win(R) technology: Immune-modulating vaccines. *Semin Immunopathol*. (2019) 41:87–95. doi: 10.1007/s00281-018-0695-8
202. Hayen SM, Otten HG, Overbeek SA, Knulst AC, Garssen J, Willemsen LEM. Exposure of intestinal epithelial cells to short- and long-chain fructo-oligosaccharides and CpG oligodeoxynucleotides enhances peanut-specific T helper 1 polarization. *Front Immunol*. (2018) 9:923. doi: 10.3389/fimmu.2018.00923
203. Seifert AM, Reiche C, Heiduk M, Tannert A, Meinecke AC, Baier S, et al. Detection of pancreatic ductal adenocarcinoma with galectin-9 serum levels. *Oncogene*. (2020) 39:3102–13. doi: 10.1038/s41388-020-1186-7
204. Cai L, Li Y, Tan J, Xu L, Li Y. Targeting LAG-3, TIM-3, and TIGIT for cancer immunotherapy. *J Hematol Oncol*. (2023) 16:101. doi: 10.1186/s13045-023-01499-1
205. Yang SL, Tan HX, Niu TT, Liu YK, Gu CJ, Li DJ, et al. The IFN-gamma-IDO1-kynurenic pathway-induced autophagy in cervical cancer cell promotes phagocytosis of macrophage. *Int J Biol Sci*. (2021) 17:339–52. doi: 10.7150/ijbs.51241
206. O'Sullivan D, van der Windt GJ, Huang SC, Curtis JD, Chang CH, Buck MD, et al. Memory CD8(+) T cells use cell-intrinsic lipolysis to support the metabolic programming necessary for development. *Immunity*. (2014) 41(1):75–88. doi: 10.1016/j.immuni.2014.06.005
207. Pham LV, Lu G, Tamayo AT, Chen J, Challagundla P, Jorgensen JL, et al. Establishment and characterization of a novel MYC/BCL2 "double-hit" diffuse large B cell lymphoma cell line, RC. *J Hematol Oncol*. (2015) 8:121. doi: 10.1186/s13045-015-0218-1
208. Tay SS, Carol H, Biro M. TriKEs and BiKEs join CARs on the cancer immunotherapy highway. *Hum Vaccin Immunother*. (2016) 12:2790–6. doi: 10.1080/21645515.2016.1198455
209. Mu-Mosley H, Ostermann L, Muftuoglu M, Vaidya A, Bonifant CL, Velasquez MP, et al. Transgenic expression of IL15 retains CD123-redirection T cells in a less differentiated state resulting in improved anti-AML activity in autologous AML PDX models. *Front Immunol*. (2022) 13:880108. doi: 10.3389/fimmu.2022.880108
210. Sullivan BN, Baggett MA, Guillory C, Jones M, Steele C. Neuroimmune responses in a new experimental animal model of cerebral aspergillosis. *mbio*. (2022) 13:e0225422. doi: 10.1128/mbio.02254-22
211. Attal M, Richardson PG, Rajkumar SV, San-Miguel J, Beksac M, Spicka I, et al. Isatuximab plus pomalidomide and low-dose dexamethasone versus pomalidomide and low-dose dexamethasone in patients with relapsed and refractory multiple myeloma (ICARIA-MM): a randomised, multicentre, open-label, phase 3 study. *Lancet*. (2019) 394:2096–107. doi: 10.1016/s0140-6736(19)32556-5
212. Weinstein B, Muresan B, Solano S, de Macedo AV, Lee Y, Su YC, et al. Efficacy and Safety of Innovative Experimental Chimeric Antigen Receptor (CAR) T-cells versus Axicabtagene ciloleucel (Yescarta) for the Treatment of Relapsed/Refractory Large B-Cell Lymphoma (LBCL): Matching Adjusted Indirect Comparisons (MAICs) and Systematic Review. *Innov Pharm*. (2021) 12(4):10.24926/iip.v12i4.4345. doi: 10.24926/iip.v12i4.4345
213. Daniele N, Scerpa MC, Caniglia M, Bernardo ME, Rossi C, Ciammetti C, et al. Transplantation in the onco-hematology field: focus on the manipulation of alphabeta and gammadelta T cells. *Pathol Res Pract*. (2012) 208:67–73. doi: 10.1016/j.prp.2011.10.006
214. Socie G. Challenges in GVHD and GVL after hematopoietic stem cell transplantation for myeloid Malignancies. *Blood*. (2025) 146:926–37. doi: 10.1182/blood.2025028617
215. Cassidy K, Martin PJ, Zeng D. Regulation of GVHD and GVL activity via PD-L1 interaction with PD-1 and CD80. *Front Immunol*. (2018) 9:3061. doi: 10.3389/fimmu.2018.03061
216. Nishimoto KP, Lamture G, Chanthery Y, Teague AG, Verma Y, Au M, et al. ADI-270: an armored allogeneic gamma delta T cell therapy designed to target CD70-expressing solid and hematologic malignancies. *J Immunother Cancer*. (2025) 13(7):e011704. doi: 10.1136/jitc-2025-011704
217. Fang X, Yan S, He L, Deng C. CAR- $\gamma\delta$ T cells: a new paradigm of programmable innate immune sentinels and their systemic applications in cancer and beyond. *Front Immunol*. (2025) 16:1735763. doi: 10.3389/fimmu.2025.1735763
218. Hou Y, Hu S, Liu C, Chen X, Wang Y, Li Y, et al. Beyond CAR-T Cells: exploring CAR-NK, CAR-M, and CAR- $\gamma\delta$ T strategies in solid tumor immunotherapy. *Front Immunol*. (2025) 16:1675807. doi: 10.3389/fimmu.2025.1675807
219. Lloret-Madrid P, Chorão P, Guerreiro M, Montesinos P. CAR-T cell therapy for acute myeloid leukemia: where do we stand now? *Curr Oncol*. (2025) 32:322. doi: 10.3390/curroncol32060322
220. Wang L, Cao ZM, Zhang LL, Dai XC, Liu ZJ, Zeng YX, et al. Helicobacter pylori and autoimmune diseases: Involving multiple systems. *Front Immunol*. (2022) 13:833424. doi: 10.3389/fimmu.2022.833424
221. Arguello Marina M, Lopez Rubio M, Castilla Garcia L. Autoimmune haemolytic anaemia. *Med Clin (Barc)*. (2023) 160:30–8. doi: 10.1016/j.medcle.2022.07.020
222. Berentsen S, Barcellini W. Autoimmune hemolytic anemias. *N Engl J Med*. (2021) 385:1407–19. doi: 10.1056/nejmra2033982
223. Watanabe N, Ikuta K, Fagaras S, Yazumi S, Chiba T, Honjo T. Migration and differentiation of autoreactive B-1 cells induced by activated gamma/delta T cells in antierythrocyte immunoglobulin transgenic mice. *J Exp Med*. (2000) 192:1577–86. doi: 10.1084/jem.192.11.1577
224. Michishita Y, Hirokawa M, Guo YM, Abe Y, Liu J, Ubukawa K, et al. Age-associated alteration of gammadelta T-cell repertoire and different profiles of activation-induced death of Vdelta1 and Vdelta2 T cells. *Int J Hematol*. (2011) 94:230–40. doi: 10.1007/s12185-011-0907-7

225. Rosso DA, Rosato M, Gomez FD, Alvarez RS, Shiromizu CM, Keitelman IA, et al. Human glomerular endothelial cells treated with Shiga toxin type 2 activate gamma delta T lymphocytes. *Front Cell Infect Microbiol.* (2021) 11:765941. doi: 10.3389/fcimb.2021.765941
226. Peters C, Kabelitz D, Wesch D. Regulatory functions of gamma delta T cells. *Cell Mol Life Sci.* (2018) 75:2125–35. doi: 10.1007/s00018-018-2788-x
227. Peterson RA. Regulatory T-cells: diverse phenotypes integral to immune homeostasis and suppression. *Toxicol Pathol.* (2012) 40:186–204. doi: 10.1177/0192623311430693
228. Kühl AA, Pawlowski NN, Grollich K, Bleszenohl M, Westermann J, Zeitz M, et al. Human peripheral $\gamma\delta$ T cells possess regulatory potential. *Immunology.* (2009) 128(4):580–8. doi: 10.1111/j.1365-2567.2009.03162.x
229. Zhang X, Chen S, Yang L, Li B, Zhu K, Li Y. The feature of TRGV and TRDV repertoire distribution and clonality in patients with immune thrombocytopenic purpura. *Hematology.* (2009) 14:237–44. doi: 10.1179/102453309x439755
230. Wang JY, Xin Y, Wang XL, Li LL, Li AM, Zhang XL. The expression pattern and role of circulating CXCR5(+) gamma delta T cells in children with newly diagnosed immune thrombocytopenia. *Front Pediatr.* (2025) 13:1646877. doi: 10.3389/fped.2025.1646877
231. Ware RE, Howard TA. Elevated numbers of gamma delta (gamma delta+) T lymphocytes in children with immune thrombocytopenic purpura. *J Clin Immunol.* (1994) 14:237–47. doi: 10.1007/bf01552310
232. Fattizzo B, Giannotta JA, Barcellini W. Mesenchymal Stem Cells in Aplastic Anemia and Myelodysplastic Syndromes: The "Seed and Soil" Crosstalk. *Int J Mol Sci.* (2020) 21(15):5438. doi: 10.3390/ijms21155438
233. Giudice V, Selleri C. Aplastic anemia: pathophysiology. *Semin Hematol.* (2022) 59:13–20. doi: 10.1053/j.seminhematol.2021.12.002
234. Younis R, Hauspurg H, Voss A. A case of hepatosplenic gamma delta T cell lymphoma with concomitant bone marrow aplasia following azathioprine therapy, treatment course and review of the literature. *Cancer Rep (Hoboken).* (2025) 8:e70334. doi: 10.1002/cnr.2.70334
235. Dyck L, Mills KHG. Immune checkpoints and their inhibition in cancer and infectious diseases. *Eur J Immunol.* (2017) 47:765–79. doi: 10.1002/eji.201646875
236. Luo M, Samandi LZ, Wang Z, Chen ZJ, Gao J. Synthetic nanovaccines for immunotherapy. *J Ctrl Rel.* (2017) 263:200–10. doi: 10.1016/j.jconrel.2017.03.033
237. Johnson CB, Zhang J, Lucas D. The role of the bone marrow microenvironment in the response to infection. *Front Immunol.* (2020) 11:585402. doi: 10.3389/fimmu.2020.585402
238. Ezech OV, Ternero-Hidalgo JJ, Lintag RMN, Han W, Yeung KL. Exploring innovations in antimicrobial protective mask filters: A review. *Adv Coll Interface Sci.* (2025) 345:103635. doi: 10.1016/j.cis.2025.103635
239. Uchil PD, Pi R, Haugh KA, Ladinsky MS, Ventura JD, Barrett BS, et al. A protective role for the lectin CD169/Siglec-1 against a pathogenic murine retrovirus. *Cell Host Microbe.* (2019) 25:87–100 e10. doi: 10.1016/j.chom.2018.11.011
240. Cho JA, Chinnappen DJF. Targeting friend and foe: Emerging therapeutics in the age of gut microbiome and disease. *J Microbiol.* (2018) 56:183–8. doi: 10.1007/s12275-018-8037-z
241. Chen ZW. Multifunctional immune responses of HMBPP-specific Vgamma2Vdelta2 T cells in M. tuberculosis and other infections. *Cell Mol Immunol.* (2013) 10:58–64. doi: 10.1038/cmi.2012.46
242. Zhang Q, Sugawara I. Immunology of tuberculosis. *World J Exp Med.* (2012) 2:70–4. doi: 10.5493/wjem.v2.i4.70
243. Shen H, Wang Y, Chen CY, Frencher J, Huang D, Yang E, et al. Th17-related cytokines contribute to recall-like expansion/effector function of HMBPP-specific Vgamma2Vdelta2 T cells after Mycobacterium tuberculosis infection or vaccination. *Eur J Immunol.* (2015) 45:442–51. doi: 10.1002/eji.201444635
244. Pauza CD, Poonia B, Li H, Cairo C, Chaudhry S. gamma delta T cells in HIV disease: Past, present, and future. *Front Immunol.* (2014) 5:687. doi: 10.3389/fimmu.2014.00687
245. Poccia F, Gougeon ML, Agrati C, Montesano C, Martini F, Pauza CD, et al. Innate T-cell immunity in HIV infection: the role of Vgamma9Vdelta2 T lymphocytes. *Curr Mol Med.* (2002) 2:769–81. doi: 10.2174/1566524023361880
246. Dawi J, Affa S, Misakyan Y, Kades S, Asim S, Olia S, et al. Dysregulation of gamma delta T-cells, CD8+ T cells, CD4+ T cells, and natural killer cells in brain pathology: Understanding HIV-induced impairment and exploring targeted GSH therapy. *Discov Med.* (2024) 36:1091–108. doi: 10.24976/discov.med.202436185.101
247. Mazzola TN, da Silva MT, Abramczuk BM, Moreno YM, Lima SC, Zorzeto TQ, et al. Impaired Bacillus Calmette-Guerin cellular immune response in HIV-exposed, uninfected infants. *AIDS.* (2011) 25:2079–87. doi: 10.1097/qad.0b013e32834bba0a
248. Rach D, Hsu HT, Nampota-Nkomba N, Mvula G, Mkandawire FA, Nyirenda OM, et al. Cord blood innate-like T cell responses in neonates born to healthy women and women living with HIV. *Front Immunol.* (2025) 16:1628145. doi: 10.3389/fimmu.2025.1628145
249. Ibraheem Y, Bayarsaikhan G, Macalinao ML, Kimura K, Yui K, Aoshi T, et al. gamma delta T cell-mediated activation of cDC1 orchestrates CD4(+) Th1 cell priming in malaria. *Front Immunol.* (2024) 15:1426316. doi: 10.3389/fimmu.2024.1426316
250. Bayarsaikhan G, Ibraheem Y, Inoue SI. $\gamma\delta$ T-cell-mediated immune responses to malaria. *Microbiol Immunol.* (2023) 67(5):239–47. doi: 10.1111/1348-0421.13059
251. Jagannathan P, Kim CC, Greenhouse B, Nankya F, Bowen K, Eccles-James I, et al. Loss and dysfunction of Vdelta2(+) gamma delta T cells are associated with clinical tolerance to malaria. *Sci Transl Med.* (2014) 6:251ra117. doi: 10.1126/scitranslmed.3009793
252. Dong P, Zhang S, Cai M, Kang N, Hu Y, Cui L, et al. Global characterization of differential gene expression profiles in mouse Vgamma1+ and Vgamma4+ gamma delta T cells. *PLoS One.* (2014) 9:e112964. doi: 10.1371/journal.pone.0112964
253. Albarrak SM, Waters WR, Stabel JR, Hostetter JM. Evaluating the cytokine profile of the WC1(+) gamma delta T cell subset in the ileum of cattle with the subclinical and clinical forms of MAP infection. *Vet Immunol Immunopathol.* (2018) 201:26–31. doi: 10.1016/j.vetimm.2018.05.003
254. Li M, Pan L, Zhang Y, Huang S, Tian Y, Tian D, et al. Regulatory TCRalpha beta (+) double negative T cells suppress gamma delta T cells and alleviate colitis. *Cell Mol Gastroenterol Hepatol.* (2025) 19:101553. doi: 10.1016/j.jcmgh.2025.101553
255. Ruggieri M, Ducasa N, Juraske C, Polo VG, Berini C, Quiroga MF, et al. Phenotypic and functional analysis of gamma delta T cells in the pathogenesis of human T-cell lymphotropic virus type 1 infection. *Front Immunol.* (2022) 13:920888. doi: 10.3389/fimmu.2022.920888
256. Mattarollo SR, Kenna T, Nieda M, Nicol AJ. Chemotherapy and zoledronate sensitize solid tumour cells to Vgamma9Vdelta2 T cell cytotoxicity. *Cancer Immunol Immunother.* (2007) 56:1285–97. doi: 10.1007/s00262-007-0279-2
257. Field KR, Wragg KM, Lee WS, Rigau M, Uldrich AP, Kent SJ, et al. Cytokines enhance human Vgamma9Vdelta2 T-cell TCR-dependent and TCR-independent effector functions. *Eur J Immunol.* (2023) 53:e2250220. doi: 10.1007/978-3-319-67546-6_5
258. Schilbach K, Frommer K, Meier S, Handgretinger R, Eyrich M. Immune response of human propagated gamma delta-T-cells to neuroblastoma recommend the Vdelta1+ subset for gamma delta-T-cell-based immunotherapy. *J Immunother.* (2008) 31:896–905. doi: 10.1097/cji.0b013e31818955ad
259. Leonard WJ, Lin JX, O'Shea JJ. The gamma(c) family of cytokines: Basic biology to therapeutic ramifications. *Immunity.* (2019) 50:832–50. doi: 10.1016/j.immuni.2019.03.028
260. Raeber ME, Zurbuchen Y, Impellizzeri D, Boyman O. The role of cytokines in T-cell memory in health and disease. *Immunol Rev.* (2018) 283:176–93. doi: 10.1111/imr.12644
261. Caramalho I, Nunes-Silva V, Pires AR, Mota C, Pinto AI, Nunes-Cabaco H, et al. Human regulatory T-cell development is dictated by Interleukin-2 and -15 expressed in a non-overlapping pattern in the thymus. *J Autoimmun.* (2015) 56:98–110. doi: 10.1016/j.jaut.2014.11.002
262. Rosser EC, Mauri C. Regulatory B cells: origin, phenotype, and function. *Immunity.* (2015) 42:607–12. doi: 10.1016/j.immuni.2015.04.005
263. Wang Y, Li X, Xing X, Xue H, Qi R, Ji H, et al. Notch-Hes1 signaling regulates IL-17A(+) gamma delta (+) T cell expression and IL-17A secretion of mouse psoriasis-like skin inflammation. *Mediators Inflammation.* (2020) 2020:8297134. doi: 10.1155/2020/8297134
264. Philippot Q, Ogishi M, Bohlen J, Puchan J, Arias AA, Nguyen T, et al. Human IL-23 is essential for IFN-gamma-dependent immunity to mycobacteria. *Sci Immunol.* (2023) 8:eabq5204. doi: 10.1016/j.clim.2023.109481
265. Ogishi M, Yang R, Rodriguez R, Golec DP, Martin E, Philippot Q, et al. Inherited human ITK deficiency impairs IFN- γ immunity and underlies tuberculosis. *J Exp Med.* (2023) 220(1):e20220484. doi: 10.1084/jem.20220484
266. Falcon RMG, Alcazar RMU, Mondragon AV, Penserga EG, Tantengco OAG. Rheumatoid arthritis and the risk of preterm birth. *Am J Reprod Immunol.* (2023) 89:e13661. doi: 10.1111/aji.13661
267. Clemente Ximenis A, Crespi Bestard C, Cambra Conejero A, Pallarés Ferreres I, Juan Mas A, Olea Vallejo JL, et al. In vitro evaluation of $\gamma\delta$ T cells regulatory function in Behçet's disease patients and healthy controls. *Hum Immunol.* (2016) 77(1):20–8. doi: 10.1016/j.humimm.2015.10.001
268. Saura-Esteller J, de Jong M, King LA, Ensing E, Winograd B, de Gruijl TD, et al. Gamma delta T-cell based cancer immunotherapy: Past-present-future. *Front Immunol.* (2022) 13:915837. doi: 10.3389/fimmu.2022.915837
269. Wang Y, Suarez ER, Kastrunes G, de Campos NSP, Abbas R, Pivetta RS, et al. Evolution of cell therapy for renal cell carcinoma. *Mol Cancer.* (2024) 23:8. doi: 10.1186/s12943-023-01911-x
270. Cowan JE, Jenkinson WE, Anderson G. Thymus medulla fosters generation of natural Treg cells, invariant gamma delta T cells, and invariant NKT cells: What we learn from intrathymic migration. *Eur J Immunol.* (2015) 45:652–60. doi: 10.1002/eji.201445108
271. Zocchi MR, Poggi A. Role of gamma delta T lymphocytes in tumor defense. *Front Biosci.* (2004) 9:2588–604. doi: 10.2741/1419

272. Parigiani MA. Gammadelta T cell expansion and their use in *in vitro* cytotoxicity assays. *Methods Cell Biol.* (2025) 193:253–64. doi: 10.1016/bs.mcb.2024.05.002
273. Barata A, Hoogland AI, Kommalapati A, Logue J, Welniak T, Hyland KA, et al. Change in patients' perceived cognition following chimeric antigen receptor T-cell therapy for lymphoma. *Transplant Cell Ther.* (2022) 28:401–e7. doi: 10.1016/j.jct.2022.05.015
274. Naiman BM, Blumerman S, Alt D, Bolin CA, Brown R, Zuerner R, et al. Evaluation of type 1 immune response in naïve and vaccinated animals following challenge with *Leptospira borgpetersenii* serovar Hardjo: involvement of WC1(+) $\gamma\delta$ and CD4 T cells. *Infect Immun.* (2002) 70(11):6147–57. doi: 10.1128/IAI.70.11.6147-6157.2002
275. Kulma I, Na-Bangchang K, Carvallo Herrera A, Ndubuisi IT, Iwasaki M, Tomono H, et al. Analysis of the Effector Functions of V δ 2 $\gamma\delta$ T Cells and NK Cells against Cholangiocarcinoma Cells. *Cells.* (2024) 13(16):1322. doi: 10.3390/cells13161322
276. Ogongo P, Steyn AJ, Karim F, Dullabh KJ, Awala I, Madansein R, et al. Differential skewing of donor-unrestricted and gammadelta T cell repertoires in tuberculosis-infected human lungs. *J Clin Invest.* (2020) 130:214–30. doi: 10.24177/po2020jci
277. Castro CD, Boughter CT, Broughton AE, Ramesh A, Adams EJ. Diversity in recognition and function of human gammadelta T cells. *Immunol Rev.* (2020) 298:134–52. doi: 10.1111/immr.12930
278. Sarrafzadeh A, Smoll N, Schaller K. Guided (VENTRI-GUIDE) versus freehand ventriculostomy: Study protocol for a randomized controlled trial. *Trials.* (2014) 15:478. doi: 10.1186/1745-6215-15-478
279. Umeyama Y, Taniguchi H, Gytoku H, Senju H, Tomono H, Takemoto S, et al. Three distinct mechanisms underlying human gammadelta T cell-mediated cytotoxicity against Malignant pleural mesothelioma. *Front Immunol.* (2023) 14:1058838. doi: 10.3389/fimmu.2023.1058838
280. Wu Z, Zhang H, Wu M, Peng G, He Y, Wan N, et al. Targeting the NKG2D/NKG2D-L axis in acute myeloid leukemia. *BioMed Pharmacother.* (2021) 137:111299. doi: 10.1016/j.biopha.2021.111299
281. Liu H, Wang S, Xin J, Wang J, Yao C, Zhang Z. Role of NKG2D and its ligands in cancer immunotherapy. *Am J Cancer Res.* (2019) 9(10):2064–78.
282. Reel PS, Reel S, Pearson E, Trucco E, Jefferson E. Using machine learning approaches for multi-omics data analysis: A review. *Biotechnol Adv.* (2021) 49:107739. doi: 10.1016/j.biotechadv.2021.107739
283. Kang M, Ko E, Mersha TB. A roadmap for multi-omics data integration using deep learning. *Brief Bioinform.* (2022) 23(1):bbab454. doi: 10.1093/bib/bbab454
284. MacEachern SJ, Forkert ND. Machine learning for precision medicine. *Genome.* (2021) 64:416–25. doi: 10.1139/gen-2020-0131
285. Petroni G, Galassi C, Gouin KH, Chen HH, Buqué A, Bloy N, et al. IL-17A-secreting $\gamma\delta$ T cells promote resistance to CDK4/CDK6 inhibitors in HR+HER2- breast cancer via CX3CR1+ macrophages. *Nat Cancer.* (2025) 6(10):1656–75. doi: 10.1038/s43018-025-01007-z
286. Laub A, Rodrigues de Almeida N, Huang S. Unconventional T cells in anti-cancer immunity. *Front Immunol.* (2025) 16:1618393. doi: 10.3389/fimmu.2025.1618393
287. Gorodetskiy V, Sidorova Y, Biderman B, Kupryshina N, Ryzhikova N, Sudarikov A. Gamma-delta T-cell large granular lymphocytic leukemia in the setting of rheumatologic diseases. *Front Cell Dev Biol.* (2024) 12:1434676. doi: 10.3389/fcell.2024.1434676
288. Sutton KS, Dasgupta A, McCarty D, Doering CB, Spencer HT. Bioengineering and serum free expansion of blood-derived gammadelta T cells. *Cytotherapy.* (2016) 18:881–92. doi: 10.1016/j.jcyt.2016.04.001
289. Nedellec S, Sabourin C, Bonneville M, Scotet E. NKG2D costimulates human V gamma 9V delta 2 T cell antitumor cytotoxicity through protein kinase C theta-dependent modulation of early TCR-induced calcium and transduction signals. *J Immunol.* (2010) 185:55–63. doi: 10.4049/jimmunol.1000373
290. Angelini DF, Zambello R, Galandrin R, Diamantini A, Placido R, Micucci F, et al. NKG2A inhibits NKG2C effector functions of gammadelta T cells: Implications in health and disease. *J Leukoc Biol.* (2011) 89:75–84. doi: 10.1189/jlb.0710413
291. Walker EM, Slisarenko N, Gerrets GL, Graspege BF, Mattison JA, Kissinger PJ, et al. Dysregulation of IL-17/IL-22 effector functions in blood and gut mucosal gamma delta T cells correlates with increase in circulating leaky gut and inflammatory markers during cART-treated chronic SIV infection in macaques. *Front Immunol.* (2021) 12:647398. doi: 10.3389/fimmu.2021.647398
292. Lameris R, Ruben JM, Iglesias-Guimaraes V, de Jong M, Veth M, van de Bovenkamp FS, et al. A bispecific T cell engager recruits both type 1 NKT and Vgamma9Vdelta2-T cells for the treatment of CD1d-expressing hematological malignancies. *Cell Rep Med.* (2023) 4:100961. doi: 10.1016/j.xcrm.2023.100961
293. Jiang S, Zheng S, Yao C, Ning D, Zou S, Zhan J, et al. Heterogeneity of gammadelta T-cell subsets and their clinical correlation in patients with AML. *Front Immunol.* (2025) 16:1552235. doi: 10.3389/fimmu.2025.1552235
294. Mathioudaki A, Wang X, Sedloev D, Huth R, Kamal A, Hundemer M, et al. The remission status of AML patients after allo-HCT is associated with a distinct single-cell bone marrow T-cell signature. *Blood.* (2024) 143:1269–81. doi: 10.1182/blood.2023021815
295. Kang SH, Hwang HJ, Yoo JW, Kim H, Choi ES, Hwang SH, et al. Expression of immune checkpoint receptors on T-cells and their ligands on leukemia blasts in childhood acute leukemia. *Anticancer Res.* (2019) 39:5531–9. doi: 10.21873/anticancer.13746
296. Arpinati M, Chirumbolo G, Urbini B, Bonifazi F, Bandini G, Saunthararajah Y, et al. Acute graft-versus-host disease and steroid treatment impair CD11c+ and CD123+ dendritic cell reconstitution after allogeneic peripheral blood stem cell transplantation. *Biol Blood Marr Transplant.* (2004) 10:106–15. doi: 10.1016/j.bbmt.2003.09.005
297. Lin J, Chen S, Zhao Z, Cummings OW, Fan R. CD123 is a useful immunohistochemical marker to facilitate diagnosis of acute graft-versus-host disease in colon. *Hum Pathol.* (2013) 44:2075–82. doi: 10.1016/j.humpath.2013.02.023
298. Boeck S, Hamann M, Pihusch V, Heller T, Diem H, Rolf B, et al. Kinetics of dendritic cell chimerism and T cell chimerism in allogeneic hematopoietic stem cell recipients. *Bone Marr Transplant.* (2006) 37:57–64. doi: 10.1038/sj.bmt.1705217
299. Barbarito G, Hiroshima L, Oppizzi L, Saini G, Kristovich K, Klein O, et al. Model-based antithymocyte globulin in alphabeta-haplo-hematopoietic stem cell transplantation facilitates engraftment, expedites T cell recovery, and mitigates the risk of acute graft-versus-host disease. *Transplant Cell Ther.* (2024) 30:810–e16. doi: 10.1016/j.jct.2024.05.015
300. Yue K, Gao H, Liang S, Wu N, Cheng C, Xu LP, et al. Improved Vdelta2(+) T cells recovery correlates to reduced incidences of mortality and relapse in acute myeloid leukemia after hematopoietic transplantation. *Ann Hematol.* (2023) 102:937–46. doi: 10.1007/s00277-023-05125-5
301. Biernacki MA, Sheth VS, Bleakley M. T cell optimization for graft-versus-leukemia responses. *JCI Insight.* (2020) 5(9):e134939. doi: 10.1172/jci.insight.134939
302. Escorcia FE, Postow MA, Barker CA. Radiotherapy and immune checkpoint blockade for melanoma: A promising combinatorial strategy in need of further investigation. *Cancer J.* (2017) 23:32–9. doi: 10.1097/ppo.0000000000000236
303. Guo Q, Zhao P, Zhang Z, Zhang J, Zhang Z, Hua Y, et al. TIM-3 blockade combined with bispecific antibody MT110 enhances the anti-tumor effect of gammadelta T cells. *Cancer Immunol Immunother.* (2020) 69:2571–87. doi: 10.1007/s00262-020-02638-0
304. Han MG, Jang BS, Kang MH, Na D, Kim IA. PI3K-gammadelta inhibitor plus radiation enhances the antitumor immune effect of PD-1 blockade in syngenic murine breast cancer and humanised patient-derived xenograft model. *Eur J Cancer.* (2021) 157:450–63. doi: 10.1016/j.ejca.2021.08.029
305. Daver N, Alotaibi AS, Bucklein V, Subklewe M. T-cell-based immunotherapy of acute myeloid leukemia: Current concepts and future developments. *Leukemia.* (2021) 35:1843–63. doi: 10.1038/s41375-021-01253-x
306. Leong SR, Sukumaran S, Hristopoulos M, Totpal K, Stainton S, Lu E, et al. An anti-CD3/anti-CLL-1 bispecific antibody for the treatment of acute myeloid leukemia. *Blood.* (2017) 129:609–18. doi: 10.1182/blood-2016-08-735365
307. Danylesko I, Shem-Tov N, Yerushalmi R, Jacoby E, Toren A, Shouval R, et al. Point of care CD19 chimeric antigen receptor (CAR) T-cells for relapsed/refractory acute myeloid leukemia (AML) with aberrant CD19 antigen expression. *Curr Res Transl Med.* (2024) 72:103471. doi: 10.1016/j.retram.2024.103471
308. Barcellini W, Fattizzo B, Corteletti A. Autoimmune hemolytic anemia, autoimmune neutropenia and aplastic anemia in the elderly. *Eur J Intern Med.* (2018) 58:77–83. doi: 10.1016/j.ejim.2018.05.034
309. Yoshida K, Murai K, Hamada H, Sato A, Miyairi Y, Wano Y. Development of severe aplastic anemia during steroid therapy for autoimmune hemolytic anemia. *Reisho Ketsueki.* (2022) 63(3):171–6. doi: 10.11406/rinketsu.63.171
310. Zhang X, Wang L, Qiao J, Wang S, Wang L, Liu L, et al. Anti-CD19 CAR-T cell therapy in relapsed/refractory t(8;21) acute myeloid leukemia with aberrant CD19 expression. *Front Immunol.* (2025) 16:1617589. doi: 10.3389/fimmu.2025.1617589
311. McMurry H, Fletcher L, Traer E. IDH inhibitors in AML-promise and pitfalls. *Curr Hematol Malig Rep.* (2021) 16:207–17. doi: 10.1007/s11899-021-00619-3
312. Kantarjian H, Kadia T, DiNardo C, Daver N, Borthakur G, Jabbour E, et al. Acute myeloid leukemia: current progress and future directions. *Blood Cancer J.* (2021) 11:41. doi: 10.1038/s41408-021-00425-3
313. Kayser S, Levis MJ. The clinical impact of the molecular landscape of acute myeloid leukemia. *Haematologica.* (2023) 108:308–20. doi: 10.3324/haematol.2022.280801
314. Sasaki H, Umezawa N, Itakura T, Iwai H, Yasuda S. Pathogenicity of functionally activated PD-1(+)CD8(+) cells and counterattacks by muscular PD-L1 through IFN-gamma in myositis. *J Autoimmun.* (2024) 142:103131. doi: 10.1016/j.jaut.2023.103131
315. Zhong H, Zou Q, Liu H, Wang X, Du S, Liang H, et al. Construction and evaluation of dual-effect cord blood natural killer cells expressing high-affinity PD-1 and chimeric antigen CD19 receptor. *Nan Fang Yi Ke Da Xue Xue Bao.* (2021) 41(12):1877–84. doi: 10.12122/j.issn.1673-4254.2021.12.18
316. Černý V, Hrdý J, Novotná O, Petrášková P, Boráková K, Kolářová L, et al. Distinct characteristics of Tregs of newborns of healthy and allergic mothers. *PLoS One.* (2018) 13(11):e0207998. doi: 10.1371/journal.pone.0207998

Glossary

HLA	Human Leukocyte Antigen	CXCL8	C-X-C Motif Chemokine Ligand 8 (Interleukin-8)
MHC	Major Histocompatibility Complex	TH1	T Helper 1 Cells
AML	Acute Myeloid Leukemia	NK	Natural Killer Cell
ALL	Acute Lymphoblastic Leukemia	GM-CSF	Granulocyte-Macrophage Colony-Stimulating Factor
CLL	Chronic Lymphocytic Leukemia	AREG	Amphiregulin
MM	Multiple Myeloma	TP53	Tumor Protein p53
AIHA	Autoimmune Hemolytic Anemia	BTK	Bruton Tyrosine Kinase
ITP	Immune thrombocytopenia	MRD	Minimal Residual Disease
AA	Aplastic Anemia	CD123	Cluster of Differentiation 123
CAR	Chimeric Antigen Receptor	CD34	Cluster of Differentiation 34
SAA	Severe Aplastic Anemia	CD19	Cluster of Differentiation 19
OS	Overall Survival	IL-13	Interleukin-13
CD4	Cluster of Differentiation 4	PD-1	Programmed Cell Death Protein 1
CD8	Cluster of Differentiation 8	TIM-3	T-cell Immunoglobulin and Mucin-domain containing-3
GMP	Good Manufacturing Practice	CR	Complete Remission
V γ 9V	T Cell Receptor Gamma 9 Delta 2 Subset	ORR	Overall Response Rate
V δ 1	T Cell Receptor Delta 1 Subset	V δ 2	T Cell Receptor Delta 2 Subset
MDS	Myelodysplastic Syndromes	BCMA	B-cell Maturation Antigen
DLBCL	Diffuse Large B-Cell Lymphoma	ADI-001	CD20 Chimeric Antigen Receptor V δ 1 γ δ
CHOP	Cyclophosphamide, Doxorubicin, Vincristine, Prednisone	HR	Hazard Ratio
IDH	Isocitrate Dehydrogenase	IDO	Indoleamine 2,3-Dioxygenase
ATG	Anti-Thymocyte Globulin	LAG-3	Lymphocyte-activation gene 3
IST	Immunosuppressive Therapy	CB-839	Telaglenastat (Glutaminase Inhibitor)
IPP	Isopentenyl Pyrophosphate	IL-2	Interleukin-2
MICA	MHC Class I Chain-related Antigen A	IL-15	Interleukin-15
HSP70	Heat Shock Protein 70	CD20	Cluster of Differentiation 20
IFN- γ	Interferon Gamma	GVHD	Graft-versus-Host Disease
TNF- α	Tumor Necrosis Factor Alpha	GVL	Graft-versus-Leukemia
NF- κ B	Nuclear Factor Kappa-light-chain-enhancer of Activated B Cells	ROR1	Receptor Tyrosine Kinase-like Orphan Receptor 1
MAPK	Mitogen-Activated Protein Kinase	HIV	Human Immunodeficiency Virus
TGF	Transforming Growth Factor Beta	IL-21	Interleukin-21
IL-6	Interleukin-6	CD27	Cluster of Differentiation 27
IL-10	Interleukin-10	GMP	Good Manufacturing Practice
IL-4	Interleukin-4	ACT	Adoptive Cell Transfer
V γ 6V	T Cell Receptor Gamma 6 Delta 1 Subset	KO	Knockout
IL	Interleukin	TCR	T Cell Receptor
TRAIL	TNF-Related Apoptosis-Inducing Ligand	HMBPP	Hydroxy-Methyl-Butenyl Pyrophosphate
ADCC	Antibody-Dependent Cell-mediated Cytotoxicity	SDHA	Succinate Dehydrogenase Complex Flavoprotein Subunit A.
CCL3	C-C Motif Chemokine Ligand 3		